Critical Design Report

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Crewed **O**perations for **P**lanetary **E**xploration

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I. Requirements Summary

The performance requirements for the COPE vehicle are summarized from the system level through the component level. Requirements are organized by Work Breakdown Structure (WBS) element and include the requirement level, identification number, requirement type, and performance requirement. The verification column identifies the subsection whose requirement assessment table evaluates each requirement, or COTS where the requirement is satisfied by the selected commercial hardware.

Table 1 System and Payload Performance Requirements

WBS	Element	L	Req. #	Type	Requirement	Verif.
1	Launch Vehicle	1	1-01	Defined	$\Delta V = 19.5\text{km/s}$	II.I
			1-02	Calculated	$M_0 = 1460000\text{kg}$	II.I
			1-03	Defined	Diameter = 6.9m	II.H
			1-04	Defined	Height = 95m	II.H
1.1	Payload	2	1.1-01	Calculated	$M_s = 5000\text{kg}$	II.I
1.1.1	Cryogenics	3	1.1.1-01	Assumed	$T = 40\text{K}$	COTS
1.1.2	Structure	3	1.1.2-01	Calculated	$V_{\text{payload}} = 1.83\text{m}^3$	COTS

Table 2 Crew Capsule Performance Requirements

WBS	Element	L	Req. #	Type	Requirement	Verif.
1.2	Crew Capsule	2	1.2-01	Calculated	$M = 26520\text{kg}$	COTS
1.2.1	ECLS	3	1.2.1-01	Defined	Supports crew of 3 astronauts	COTS
			1.2.1-02	Assumed	Mission duration $\geq$ 14days	COTS
1.2.2	Thermal Control	3	1.2.2-01	Assumed	$T_{\text{cabin}} = 297\text{K}$	COTS
			1.2.2-02	Assumed	$291\text{K} \leq T_{\text{cabin}} \leq 300\text{K}$	COTS
			1.2.2-03	Assumed	$Q_{\text{cool}} = 5\text{kW}$	COTS
1.2.3	Propulsion	3	1.2.3-01	Calculated	$F_{\text{abort}} \geq 1040\text{kN}$	COTS
1.2.4	Structure	3	1.2.4-01	Calculated	$M_s = 9000\text{kg}$	COTS
1.2.5	Electrical Systems	3	1.2.5-01	Assumed	$V_{\text{bus}} = 28\text{VDC}$	COTS
			1.2.5-02	Assumed	$P_{\text{cont}} = 20\text{kW}$	COTS
			1.2.5-03	Assumed	$P_{\text{peak}} = 35\text{kW}$	COTS
			1.2.5-04	Assumed	$E_{\text{batt}} = 72\text{hr}$	COTS
1.2.6	GNC	3	1.2.6-01	Assumed	$F_{\text{RCS}} \leq 111\text{N}$	COTS
1.2.7	Safety Systems	3	1.2.7-01	Assumed	$a_{\text{abort,max}} < 4g$	COTS
			1.2.7-02	Assumed	$t_{\text{abort}} \leq 2\text{s}$	COTS
			1.2.7-03	Assumed	$P_{\text{survival}} \geq 0.99$	COTS
			1.2.7-04	Assumed	$t_{\text{safe config}} \leq 24\text{hr}$	COTS

Table 3 Stage 1 Performance Requirements

WBS	Element	L	Req. #	Type	Requirement	Verif.
1.3	Stage 1 Vehicle	2	1.3-01	Allocated	$\Delta V = 3650\text{m/s}$	II.I
			1.3-02	Allocated	$M_i = 820000\text{kg}$	II.I
			1.3-03	Allocated	$M_0 = 1460000\text{kg}$	II.I
			1.3-04	Calculated	$T/W = 1.4$	II.A
1.3.1	Propulsion	3	1.3.1-01	Allocated	$\Delta V = 3.6\text{km/s}$	II.I
			1.3.1-02	Allocated	$M_{\text{ox}} = 737000\text{kg}$	II.I
			1.3.1-03	Allocated	$M_f = 211000\text{kg}$	II.I
			1.3.1-04	Calculated	$F = 2600\text{kN}$	II.A
1.3.1.1	Engine	4	1.3.1.1-01	Defined	$P_c = 25.53\text{MPa}$	II.A
			1.3.1.1-02	Defined	$A_t = 0.0550\text{m}^2$	II.A
			1.3.1.1-03	Defined	$\epsilon = 40$	II.A
			1.3.1.1-04	Defined	$D_e = 1.674\text{m}$	II.B
			1.3.1.1-05	Calculated	$I_{sp,SL} \geq 329\text{s}$	II.A

WBS	Element	L	Req. #	Type	Requirement	Verif.
			1.3.1.1-06	Calculated	$I_{sp,vac} \geq 358$s	II.A
			1.3.1.1-07	Calculated	$\dot{m}_{eng} \geq 771.5$kg/s	II.A
			1.3.1.1-08	Defined	$T_{wg,max} \leq 1000$K	II.C
			1.3.1.1-09	Allocated	$m_{eng} \leq 2800$kg	II.I
1.3.1.2	Turbomachinery, Feed & Pressurization	4	1.3.1.2-01	Defined	$v_{\text{feed,max}} \leq 10$m/s	II.F
			1.3.1.2-02	Calculated	$D_{\text{LOX}} \geq 259$mm	II.F
			1.3.1.2-03	Calculated	$D_{\text{CH}_4} \geq 227$mm	II.F
			1.3.1.2-04	Calculated	$P_{\text{in,LOX}} \geq 0.351$MPa	II.F
			1.3.1.2-05	Calculated	$P_{\text{in,CH}_4} \geq 0.285$MPa	II.F
			1.3.1.2-06	Defined	$\eta_{\text{pump,LOX}} \geq 0.80$	II.E
			1.3.1.2-07	Defined	$\eta_{\text{pump,CH}_4} \geq 0.80$	II.E
			1.3.1.2-08	Derived	$P_{\text{turb}} \geq 1.10 P_{\text{pump}}$	II.D
			1.3.1.2-09	Derived	$\text{NPSH}_\text{A}/\text{NPSH}_\text{C} \geq 1.10$	II.E
1.3.1.3	Thrust Vector Control	4	1.3.1.3-01	Defined	$\lvert\delta_{\max}\rvert \geq 6°$	II.G
			1.3.1.3-02	Derived	$M_{\text{TVC}} \geq 1.5 M_{\text{disturbance}}$	II.G
			1.3.1.3-03	Calculated	$F_{\text{act,applied}} \leq 160.4$kN	II.G
1.3.2	Structure	3	1.3.2-01	Calculated	$M_s = 154000$kg	II.I
			1.3.2-02	Calculated	$H_{\text{ox}} = 18.35$m	II.H
			1.3.2-03	Calculated	$H_{\text{fuel}} = 14.17$m	II.H
1.3.2.1	LOX Tank	4	1.3.2.1-01	Calculated	$V_{\text{LOX}} \geq 665$ m^3	II.H
1.3.2.2	CH_4 Tank	4	1.3.2.2-01	Calculated	$V_{\text{CH}_4} \geq 514$ m^3	II.H
1.3.2.3	Interstage	4	1.3.2.3-01	Calculated	$F_{\text{axial,max}} \leq 15.5$MN	II.H
			1.3.2.3-02	Calculated	$\sigma_{\text{axial,max}} \leq 180$MPa	II.H
			1.3.2.3-03	Calculated	$L_{\text{interstage}} \geq 4.7$m	II.H
1.3.3	Thermal Control	3	1.3.3-01	Assumed	$T_{\text{LOX}} = 90$K	COTS
			1.3.3-02	Assumed	$T_{\text{CH}_4} = 111$K	COTS
			1.3.3-03	Assumed	$T_{\text{max,avionics}} \leq 323$K	COTS
			1.3.3-04	Assumed	$t_{\text{MLI}} = 25$mm	COTS
1.3.5	Stage Separation	3	1.3.5-01	Assumed	Separation pressure ≥ 3MPa	COTS
			1.3.5-02	Assumed	Separation velocity ≥ 1.5m/s	COTS
			1.3.5-03	Assumed	Separation time ≤ 0.5s	COTS
1.3.6	GNC	3	1.3.6-01	Assumed	$F_{\text{RCS}} \leq 111$N	COTS
1.3.7	Safety Systems	3	1.3.7-01	Assumed	$P_{\text{fire detect}} \geq 0.95$	COTS

Table 4 Stage 2 Performance Requirements

WBS	Element	L	Req. #	Type	Requirement	Verif.
1.4	Stage 2 Vehicle	2	1.4-01	Allocated	$\Delta V = 3650$m/s	II.I
			1.4-02	Allocated	$M_i = 82000$kg	II.I
			1.4-03	Allocated	$M_0 = 356699$kg	II.I
			1.4-04	Calculated	$T/W > 1$	II.A
			1.4-05	Calculated	$F = 2980$kN	II.A
1.4.1	Propulsion Pre-Refueling	3	1.4.1-01	Allocated	$\Delta V = 3.6$km/s	II.I
			1.4.1-02	Allocated	$M_{\text{ox}} = 228000$kg	II.I
			1.4.1-03	Allocated	$M_{\text{pr}} = 65000$kg	II.I
1.4.1	Propulsion Post-Refueling	3	1.4.1-04	Allocated	$\Delta V = 9.5$km/s	II.I
			1.4.1-05	Allocated	$M_{\text{ox}} = 588000$kg	II.I
			1.4.1-06	Allocated	$M_{\text{pr}} = 168000$kg	II.I
1.4.1.1	Engine	4	1.4.1.1-01	Defined	$P_c = 26.08$MPa	II.A
			1.4.1.1-02	Defined	$A_t = 0.0550$m^2	II.A
			1.4.1.1-03	Defined	$\epsilon = 100$	II.A
			1.4.1.1-04	Defined	$D_e = 2.646$m	II.B
			1.4.1.1-05	Calculated	$I_{sp,vac} \geq 372.6$s	II.A
			1.4.1.1-06	Calculated	$\dot{m}_{eng} \geq 767.3$kg/s	II.A
			1.4.1.1-07	Defined	$T_{wg,max} \leq 1000$K	II.C

WBS	Element	L	Req. #	Type	Requirement	Verif.
			1.4.1.1-08	Allocated	$m_{eng} \leq 3300\text{kg}$	II.I
1.4.1.2	Turbomachinery, Feed & Pressurization	4	1.4.1.2-01	Defined	$v_{\text{feed,max}} \leq 10\text{m/s}$	II.F
			1.4.1.2-02	Calculated	$D_{\text{LOX}} \geq 262\text{mm}$	II.F
			1.4.1.2-03	Calculated	$D_{\text{CH}_4} \geq 230\text{mm}$	II.F
			1.4.1.2-04	Calculated	$P_{\text{in,LOX}} \geq 0.351\text{MPa}$	II.F
			1.4.1.2-05	Calculated	$P_{\text{in,CH}_4} \geq 0.285\text{MPa}$	II.F
			1.4.1.2-06	Defined	$\eta_{\text{pump,LOX}} \geq 0.80$	II.E
			1.4.1.2-07	Defined	$\eta_{\text{pump,CH}_4} \geq 0.80$	II.E
			1.4.1.2-08	Derived	$P_{\text{turb}} \geq 1.10 P_{\text{pump}}$	II.D
			1.4.1.2-09	Derived	$\text{NPSH}_\text{A}/\text{NPSH}_\text{C} \geq 1.10$	II.E
1.4.1.3	Thrust Vector Control	4	1.4.1.3-01	Defined	$\lvert\delta_{\text{max}}\rvert \geq 6°$	II.G
			1.4.1.3-02	Derived	$M_{\text{TVC}} \geq 1.5 M_{\text{disturbance}}$	II.G
			1.4.1.3-03	Calculated	$F_{\text{act,applied}} \leq 256.5\text{kN}$	II.G
1.4.2	Structure	3	1.4.2-01	Allocated	$M_s = 32500\text{kg}$	II.I
			1.4.2-02	Calculated	$H_{\text{ox tank}} = 15.25\text{m}$	II.H
			1.4.2-03	Calculated	$H_{\text{fuel tank}} = 11.78\text{m}$	II.H
1.4.2.1	LOX Tank	4	1.4.2.1-01	Calculated	$V_{\text{LOX}} \geq 568\ \text{m}^3$	II.H
1.4.2.2	CH_4 Tank	4	1.4.2.2-01	Calculated	$V_{\text{CH}_4} \geq 439\ \text{m}^3$	II.H
1.4.2.3	Capsule Adapter	4	1.4.2.3-01	Calculated	$F_{\text{axial,max}} \leq 1.4\text{MN}$	II.H
			1.4.2.3-02	Calculated	$\sigma_{\text{meridional,max}} \leq 100\text{MPa}$	II.H
1.4.3	Thermal Control	3	1.4.3-01	Assumed	$T_{\text{LOX}} = 90\text{K}$	COTS
			1.4.3-02	Assumed	$T_{\text{CH}_4} = 111\text{K}$	COTS
			1.4.3-03	Assumed	$T_{\text{max,avionics}} = 323\text{K}$	COTS
			1.4.3-04	Assumed	$t_{\text{MLI}} = 40\text{mm}$	COTS
1.4.4	Electrical Systems	3	1.4.4-01	Assumed	$V_{\text{bus}} = 28\text{VDC}$	COTS
			1.4.4-02	Assumed	$P_{\text{cont}} = 80\text{kW}$	COTS
			1.4.4-03	Assumed	$P_{\text{peak}} = 140\text{kW}$	COTS
			1.4.4-04	Assumed	$E_{\text{batt}} = 75\text{kWh}$	COTS
1.4.5	Refueling Systems	3	1.4.5-01	Assumed	$\dot{M}_f = 100\text{kg/s}$	COTS
1.4.6	GNC	3	1.4.6-01	Assumed	$F_{\text{RCS}} = 111\text{N}$	COTS
1.4.7	Safety Systems	3	1.4.7-01	Assumed	$P_{\text{FD}} = 0.99$	COTS
			1.4.7-02	Defined	$t_{\text{fault isolation}} \leq 1\text{s}$	COTS
			1.4.7-03	Assumed	$t_{\text{safe haven}} = 24\text{hr}$	COTS

Table 5 Refueling Vehicle and Ground Systems Performance Requirements

WBS	Element	L	Req. #	Type	Requirement	Verif.
1.5	Refueling Vehicle	2	1.5-01	Allocated	$M_0 = 356699\text{kg}$	II.I
1.5.1	Propulsion	3	1.5.1-01	Defined	$\Delta V = 3.6\text{km/s}$	II.I
			1.5.1-02	Calculated	$M_{\text{ox}} \leq 252141\text{kg}$	II.H
			1.5.1-03	Calculated	$M_f \leq 72040\text{kg}$	II.H
			1.5.1-04	Calculated	$F = 2750\text{kN}$	II.A
1.5.2	Structure	3	1.5.2-01	Calculated	$M_s = 32500\text{kg}$	II.I
			1.5.2-02	Calculated	$H_{\text{ox}} \leq 18.35\text{m}$	II.H
			1.5.2-03	Calculated	$H_{\text{fuel}} \leq 14.17\text{m}$	II.H
1.5.3	Thermal Control	3	1.5.3-01	Assumed	$T_{\text{LOX}} = 90\text{K}$	COTS
			1.5.3-02	Assumed	$T_{\text{CH}_4} = 111\text{K}$	COTS
			1.5.3-03	Assumed	$T_{\text{max,avionics}} \leq 323\text{K}$	COTS
			1.5.3-04	Assumed	$t_{\text{MLI}} = 25\text{mm}$	COTS
1.5.4	Electrical Systems	3	1.5.4-01	Defined	$V_{\text{bus}} = 28\text{VDC}$	COTS
			1.5.4-02	Calculated	$P_{\text{cont}} = 40\text{kW}$	COTS
			1.5.4-03	Calculated	$P_{\text{peak}} = 80\text{kW}$	COTS
			1.5.4-04	Assumed	$E_{\text{batt}} \geq 24\text{hr}$	COTS
1.5.5	Refueling Systems	3	1.5.5-01	Assumed	$\dot{M}_f = 100\text{kg/s}$	COTS
1.5.6	GNC	3	1.5.6-01	Assumed	$F_{\text{RCS}} \leq 111\text{N}$	COTS

WBS	Element	L	Req. #	Type	Requirement	Verif.
			1.5.6-02	Assumed	Docking accuracy ≤ 0.1m	COTS
1.5.7	Safety Systems	3	1.5.7-01	Assumed	$P_{\text{leak detect}} \geq 0.99$	COTS
			1.5.7-02	Assumed	Emergency shutdown ≤ 1s	COTS
1.6	Ground Systems	2	1.6-01	Defined	Supports vertical launch operations	COTS
			1.6-02	Assumed	Pad structural load capacity ≥ 1.5× vehicle weight	COTS

II. Engineering Analyses

Engineering analyses were conducted to evaluate the detailed design of the COPE launch vehicle and verify the performance requirements established in Section I. The following sections evaluate the propulsion, thermal, feed, thrust vector control, structural, and vehicle level performance of the design. Where applicable, analytical results are compared with existing systems and the requirements established for the COPE vehicle.

A. Engine Operating Point and Performance

The Stage 1 and Stage 2 engines use liquid oxygen (LOX) and liquid methane (CH_4) in a full-flow staged-combustion (FFSC) cycle. Both engines use a common throat area of 0.0550 m^2. The Stage 1 engine operates at a chamber pressure of 25.53 MPa with an expansion ratio of 40 and an oxidizer-to-fuel ratio of 3.50. The Stage 2 engine operates at a chamber pressure of 26.08 MPa with an expansion ratio of 100 and an oxidizer-to-fuel ratio of 3.50.

NASA CEA was used to determine the equilibrium combustion properties and theoretical performance of the LOX/CH_4 propellant combination. A characteristic-velocity efficiency of $\eta_{c^*} = 0.98$ was applied to account for combustion losses, while the nozzle efficiency η_{C_f} was obtained from the nozzle analysis in Subsection II.B.

The resulting operating conditions and delivered engine performance are summarized in Table 6.

Table 6 COPE Engine Operating Point and Performance

Parameter	Stage 1	Stage 2
Chamber pressure, P_c [MPa]	25.53	26.08
Throat area, A_t [m^2]	0.0550	0.0550
Expansion ratio, ϵ	40	100
O/F ratio	3.50	3.50
Chamber temperature, T_c [K]	3733	3736
Molecular weight [kg/kmol]	22.32	22.32
Throat specific heat ratio, γ	1.132	1.132
Ideal characteristic velocity, c^* [m/s]	1856.3	1856.7
Delivered characteristic velocity, c^* [m/s]	1819.2	1819.6
Vacuum I_{sp}, equilibrium [s]	370.9	386.5
Vacuum I_{sp}, frozen [s]	348.9	360.6
Sea-level I_{sp}, equilibrium [s]	340.9	–
Nozzle efficiency, η_{C_f}	0.9921	0.9919
Delivered vacuum I_{sp} [s]	360.6	375.7
Delivered sea-level I_{sp} [s]	331.2	–
Delivered mass flow rate [kg/s]	771.9	788.3
Delivered vacuum thrust [MN]	2.730	2.905
Delivered sea-level thrust [MN]	2.507	–

The COPE engine operating points were compared with existing staged-combustion rocket engines to determine whether the selected chamber pressures, thrust levels, and specific impulse values are reasonable for a high-performance liquid rocket engine. The comparison is shown in Table 7.

Table 7 Comparison with Existing Staged-Combustion Engines

Parameter	**COPE S1**	**COPE S2**	**Raptor 2** [20]	**RS-25** [21]	**RD-180** [22]
Propellant	LOX/CH_4	LOX/CH_4	LOX/CH_4	LOX/LH_2	LOX/RP-1
Cycle	FFSC	FFSC	FFSC	Staged (fuel-rich)	Staged (ox-rich)
P_c [MPa]	25.53	26.08	30.0	20.64	25.75
Sea-level thrust [kN]	2507	–	2256	1859	3827
Vacuum thrust [kN]	2730	2905	–	2279	4150
Sea-level I_{sp} [s]	331	–	327	369*	311
Vacuum I_{sp} [s]	361	376	–	452	337*
Expansion ratio	40	100	–	69	36.4
Engine T/W	95	95	144	54	74

The comparison shows that the COPE chamber pressures and thrust levels are within the range of existing high-performance staged-combustion engines. Differences in specific impulse are primarily caused by the propellant combination, expansion ratio, and engine operating conditions.

Using the post-refueling mission mass ratio and velocity budget gives a minimum Stage 2 vacuum specific impulse of 372.6 s. The delivered value of 375.7 s exceeds this requirement and provides approximately 80 m/s of mission velocity reserve.

Table 8 summarizes the requirements directly evaluated by the engine operating-point analysis.

Table 8 Engine Operating Point Requirement Assessment

Req. #	**Requirement**	**Analysis Result**	**Status**
1.3.1.1-01/02/03	P_c = 25.53 MPa, A_t = 0.0550 m^2, ϵ = 40	Stage 1 design operating point	Met
1.4.1.1-01/02/03	P_c = 26.08 MPa, A_t = 0.0550 m^2, ϵ = 100	Stage 2 design operating point	Met
1.3.1.1-05	$I_{sp,SL} \geq 329$ s	331.2 s	Met
1.3.1.1-06	$I_{sp,vac} \geq 358$ s	360.6 s	Met
1.3.1.1-07	$\dot{m}_{eng} \geq 771.5$ kg/s	771.9 kg/s	Met
1.4.1.1-05	$I_{sp,vac} \geq 372.6$ s	375.7 s	Met
1.4.1.1-06	$\dot{m}_{eng} \geq 767.3$ kg/s	788.3 kg/s	Met
1.3-04	$T/W = 1.4$	1.43 at current sized mass	Met
1.3.1-04	F = 2600 kN	2507 kN SL; vehicle T/W requirement remains satisfied	Met (Within 5%)
1.4-04	$T/W > 1$	1.04 at post-refuel TLI ignition; 2.49 on ascent	Met
1.4-05	F = 2980 kN	2905 kN vac (−2.5%)	Met (Within 5%)
1.5.1-04	F = 2750 kN	2905 kN vac; refueling vehicle uses the Stage 2 engine	Met

The Stage 1 engine produces 2507 kN of sea-level thrust, which is below the original 2600 kN requirement. However, the vehicle still satisfies the Stage 1 thrust-to-weight requirement, achieving $T/W = 1.401$ using the PDR vehicle mass and $T/W = 1.43$ using the current sized vehicle mass. Therefore, the individual Stage 1 thrust requirement remains met within tolerances, while the higher-level vehicle thrust-to-weight requirement is satisfied.

The Stage 2 engine produces 2905 kN in vacuum, which is 2.5% below the original 2980 kN requirement. The Stage 2 vehicle nevertheless maintains $T/W = 1.04$ at the critical post-refueling TLI ignition condition and $T/W = 2.49$ during ascent. Therefore, the individual Stage 2 thrust requirement remains met within tolerances, while the vehicle-level thrust-to-weight requirement is satisfied.

B. Nozzle Contour and CFD

The Stage 1 and Stage 2 nozzles were designed using an 80% Rao bell contour. Both engines use the common throat geometry established in Subsection II.A, with a contraction ratio of 2.5, characteristic length of $L^* = 1.10$ m, and a 30° converging section. Throat radii of curvature of $1.5R_t$ and $0.382R_t$ were used, and the initial and exit nozzle angles were selected using the Rao bell design method [1].

The resulting Stage 1 and Stage 2 thrust chamber geometries are summarized in Table 9.

Table 9 Stage 1 and Stage 2 Thrust Chamber Geometry

Parameter	Stage 1	Stage 2
Expansion ratio, ϵ	40	100
Throat radius, R_t [mm]	132.3	132.3
Chamber radius, R_c [mm]	209.2	209.2
Cylinder length [mm]	276	276
Injector-to-throat length [mm]	516	516
Initial nozzle angle, θ_n	32.5°	36.5°
Exit angle, θ_e	7.5°	5.5°
Divergence factor, λ	0.970	0.967
Bell length [mm]	2109	3561
Exit diameter, D_e [m]	1.674	2.646
Overall thrust chamber length [m]	2.62	4.08
Wetted area [m^2]	8.5	21.7
Exit Mach number, M_e	4.19	4.90

Computational fluid dynamics (CFD) was used to evaluate the nozzle flow and compare the resulting performance with the NASA CEA predictions. Each nozzle was modeled as an inviscid, axisymmetric flow field using the combustion conditions established in Subsection II.A. The CFD solution was initialized using the CEA throat conditions and evaluated until the predicted thrust reached a steady value.

The resulting CFD performance is compared with the CEA predictions in Table 10. The CFD solution predicts slightly lower thrust and thrust coefficient than CEA due to losses associated with the nozzle flow field. The ratio between the CFD and CEA thrust coefficients is used as the nozzle efficiency in Subsection II.A.

Table 10 Nozzle CFD Comparison with CEA Results

Parameter	S1 CFD	S1 CEA	S2 CFD	S2 CEA
Mass flow rate [kg/s]	759.6	756.8	775.0	772.9
Vacuum thrust [MN]	2.730	2.752	2.915	2.929
Vacuum thrust coefficient, $C_{f,vac}$	1.9439	1.9597	2.0319	2.0417
Vacuum I_{sp} (ideal c^*) [s]	366.4	370.8	383.5	386.4
Exit pressure [kPa]	70.5	64.6	24.0	20.0

The difference between the CFD and CEA exit pressures is attributed to the different gas models: the CFD treats the exhaust as a non-reacting gas, while CEA assumes chemical equilibrium throughout the expansion. For Stage 1, the CFD exit pressure gives a sea-level pressure ratio of $p_e/p_a = 70.5/101.3 = 0.70$ (0.64 using the CEA exit pressure). Both values exceed the minimum ratio of 0.40 used to screen for nozzle flow separation [2]. Therefore, the Stage 1 nozzle is not expected to experience flow separation at sea level. Additional nozzle and plume CFD results are provided in Appendix A.1.

Table 11 Nozzle Contour and CFD Requirement Assessment

Req. #	Requirement	Analysis Result	Status
1.3.1.1-04	$D_e = 1.674$ m	$D_e = 1.674$ m	Met
1.4.1.1-04	$D_e = 2.646$ m	$D_e = 2.646$ m	Met

C. Thermal Analysis and Regenerative Cooling

The thrust chamber and nozzle experience high thermal loads from the combustion gases and require cooling to maintain the chamber liner below its allowable temperature. Regenerative cooling was evaluated using methane as the coolant. The thermal model calculates local wall temperature, heat flux, coolant heating, and pressure loss along the chamber and nozzle contour. These results are coupled to the engine cycle to account for the heated methane supplied to the preburners and the pressure required upstream of the cooling jacket.

The gas-side convective heat flux is determined from

$$q'' = h_g(T_{aw} - T_{wg}), \tag{1}$$

where h_g is the gas-side heat-transfer coefficient, T_{aw} is the adiabatic wall temperature, and T_{wg} is the hot-gas-side wall temperature. The gas-side heat-transfer coefficient is estimated using the Bartz correlation with a wall-temperature correction.

Heat transfer through the chamber liner is determined from

$$q'' = \frac{k_w(T_{wg} - T_{wc})}{t_w}, \tag{2}$$

where k_w is the wall thermal conductivity, T_{wc} is the coolant-side wall temperature, and t_w is the hot-wall thickness.

For each rectangular cooling channel, the hydraulic diameter is

$$D_h = \frac{2wd}{w + d}, \tag{3}$$

where w and d are the local channel width and depth. The coolant-side heat-transfer coefficient is calculated using

$$Nu = 0.023Re^{0.8}Pr^{0.4}, \qquad h_c = \frac{Nu\, k_c}{D_h}. \tag{4}$$

Methane thermophysical properties are obtained by linear interpolation of a CoolProp HEOS property table covering 110-625 K and 30-60 MPa. The model checks that $Re \geq 10^4$ throughout the cooling passages.

Methane enters each cooling branch at 120 K. The Stage 1 cooling inlet pressure is fixed at the methane pump discharge pressure of 51.2 MPa. For Stage 2, the cooling inlet pressure is updated iteratively based on the calculated cooling pressure loss and preburner pressure requirements.

The cooling jacket is divided into three parallel branches. The chamber branch receives 35% of the methane flow and travels from the throat toward the injector. The inner-nozzle branch receives 30% and travels from the throat to an area ratio of $A/A_t = 10$. The extension branch receives the remaining 35% and travels from the nozzle exit toward the same area-ratio boundary. The channel width increases with local radius, while the channel depth is held constant within each branch. The cooling geometry and model inputs are summarized in Table 12.

Table 12 Regenerative Cooling Geometry and Model Inputs

Parameter	Stage 1	Stage 2
Number of cooling channels	1600	1600
Channel width at throat [mm]	0.300	0.300
Chamber channel depth [mm]	6.80	6.80
Inner-nozzle channel depth [mm]	5.00	5.00
Extension channel depth [mm]	5.00	5.00
Hot-wall thickness [mm]	0.280	0.280
Coolant inlet temperature [K]	120	120
Coolant inlet pressure [MPa]	51.2	Iteratively calculated
Chamber / inner-nozzle / extension flow split [%]	35 / 30 / 35	35 / 30 / 35
Inner-nozzle / extension boundary, A/A_t	10	10
Wall thermal conductivity [W/(m·K)]	300	300
Wall temperature limit [K]	1000	1000
Channel roughness [μm]	1.0	1.0
Inlet / outlet minor-loss coefficients	1.0 / 1.0	1.0 / 1.0
Downstream pressure allowance [MPa]	2.75	2.75

The chamber and nozzle contour is discretized using a nominal 1200-segment mesh with additional points at geometry and cooling-branch boundaries. Local gas Mach number is calculated from the isentropic area-Mach relation, and the adiabatic wall temperature is determined using a recovery factor of $Pr_g^{1/3}$.

Heat transfer through the channel ribs is included using a fin-efficiency correction. Gas-side convection, wall conduction, and coolant-side convection are solved together to determine the local heat flux and wall temperatures. Coolant heating is tracked through an enthalpy balance along each cooling branch,

$$h_{i+1} = h_i + \frac{q_i'' \, 2\pi r_i \Delta s_i}{\dot{m}_b}, \tag{5}$$

where $\dot{m}_b$ is the branch mass flow rate and Δs_i is the segment length along the wall. The updated coolant pressure and enthalpy are then used to determine the temperature and thermophysical properties for the next segment.

Cooling pressure changes include wall friction, coolant acceleration, and inlet and outlet minor losses. The Darcy friction factor is calculated using the Haaland correlation, with the segment friction loss calculated from

$$\Delta P_{f,i} = f_{D,i} \frac{\Delta s_i}{D_{h,i}} \frac{G_i^2}{2\rho_i}, \tag{6}$$

where G_i is the coolant mass flux. The three branch outlets are pressure-balanced before the heated methane is recombined and supplied to the engine cycle.

For Stage 2, the required methane pump discharge pressure is updated using

$$P_{d,\mathrm{CH_4}} = \max\left(P_{\mathrm{PB,FR}}, P_{\mathrm{PB,OR}}\right) + \Delta P_{\mathrm{regen}} + \Delta P_{\mathrm{allow}}, \tag{7}$$

where $\Delta P_{\mathrm{regen}}$ is the calculated cooling-jacket pressure loss and $\Delta P_{\mathrm{allow}} = 2.75$ MPa accounts for downstream line and injector losses.

A material trade study was performed to select the chamber liner material. GRCop-42, CuCrZr, NASA HR-1, and Inconel 718 were compared based on thermal conductivity, temperature capability, manufacturability, durability, and heritage. Thermal conductivity and temperature capability were given the highest weights because the chamber liner must efficiently transfer heat into the regenerative coolant while remaining below its allowable operating temperature. The resulting trade study is shown in Table 13.

Table 13 Combustion Chamber Liner Material Trade Study

Criteria	Weight	GRCop-42	CuCrZr	NASA HR-1	Inconel 718
Thermal Conductivity	0.30	5.0	5.0	2.0	2.0
Temperature Capability	0.30	4.0	2.5	4.0	4.5
Manufacturability	0.15	3.5	4.5	3.5	5.0
Durability	0.15	4.5	3.5	4.5	4.5
Heritage	0.10	4.5	4.0	4.0	5.0
Total	**1.00**	**4.35**	**3.85**	**3.40**	**3.88**

GRCop-42 received the highest weighted score and was selected as the chamber liner material. Its high thermal conductivity and temperature capability make it well suited for transferring heat from the combustion gases into the regenerative coolant while maintaining the liner below its design temperature limit [6, 7]. CuCrZr provides comparable thermal conductivity but has a lower allowable operating temperature [11]. Inconel 718 provides greater high-temperature capability and NASA HR-1 provides comparable capability, but both have substantially lower thermal conductivity, reducing their effectiveness as regeneratively cooled chamber liner materials [8, 9, 12]. GRCop-42 was therefore carried forward in the final thermal model using a thermal conductivity of 300 $\mathrm{W/(m \cdot K)}$ and a design temperature limit of 1000 K.

The final regenerative cooling design was evaluated using the selected GRCop-42 liner and the cooling geometry defined in Table 12. The resulting thermal and pressure performance is summarized in Table 14.

Table 14 Regenerative Cooling Design and Performance

Parameter	Stage 1	Stage 2
Peak gas-side heat-transfer coefficient [W/(m^2·K)]	49140	49946
Peak heat flux [MW/m^2]	144.1	146.3
Peak hot-wall temperature [K]	797	801
Throat hot-wall temperature [K]	795	800
Throat coolant velocity [m/s]	41	42
Cooling jacket pressure drop [MPa]	5.14	5.36
Cooling jacket outlet pressure [MPa]	46.06	45.54
Required outlet pressure [MPa]	44.62	45.54
Mixed coolant outlet temperature [K]	381	412
Total heat load [MW]	144	165
Regeneratively cooled area [m^2]	8.5	21.7

The nominal analysis predicts maximum hot-wall temperatures of 797 K for Stage 1 and 801 K for Stage 2, providing thermal margins of 203 K and 199 K, respectively, relative to the 1000 K design limit. The Stage 1 cooling-jacket outlet pressure of 46.06 MPa exceeds the required 44.62 MPa by 1.44 MPa. For Stage 2, the pump discharge pressure was solved iteratively with the cooling model, resulting in a cooling-jacket outlet pressure of 45.54 MPa, which meets the required 45.54 MPa.

The calculated maximum wall temperatures were compared with the Stage 1 and Stage 2 engine thermal requirements in Table 15.

Table 15 Regenerative Cooling Requirement Assessment

Req. #	Requirement	Analysis Result	Status
1.3.1.1-08	$T_{wg,\max} \leq 1000$ K	797 K	Met
1.4.1.1-07	$T_{wg,\max} \leq 1000$ K	801 K	Met

D. Engine Cycle Balance

A full-flow staged-combustion (FFSC) cycle was evaluated to determine the pressure and power requirements throughout the Stage 1 and Stage 2 propulsion systems. The analysis includes the main propellant pumps, regenerative cooling system described in Subsection II.C, fuel-rich and oxidizer-rich preburners, turbines, and main injector. A main injector pressure drop equal to 20% of chamber pressure was used, resulting in a turbine discharge pressure of $1.20P_c$. Pump and turbine efficiencies of 0.80 and 0.85 were assumed, respectively, along with a mechanical efficiency of 0.98.

The fuel-rich and oxidizer-rich preburner mixture ratios were set to 0.25 and 50, respectively. Preburner gas properties were obtained from NASA CEA. The Stage 1 pump discharge pressures were established at 51.20 MPa on the fuel side and 45.40 MPa on the oxidizer side. The preburner pressure on each side was then solved such that the turbine provides at least 10% more power than required by its corresponding pump.

For Stage 2, the pump discharge pressures were solved iteratively with the regenerative cooling system. The fuel pump discharge pressure accounts for the preburner pressure, cooling-jacket pressure loss, and downstream pressure allowance established in Subsection II.C. This resulted in Stage 2 pump discharge pressures of 50.90 MPa on the fuel side and 45.54 MPa on the oxidizer side.

The turbine must provide sufficient power to drive the corresponding pump while maintaining a design power margin. Therefore, the cycle was solved to satisfy

$$P_{\text{turb}} \geq 1.10 P_{\text{pump}}. \tag{8}$$

The regenerative cooling system also increases the methane temperature and enthalpy before it enters the preburners. This increase was included in the cycle balance to determine the resulting preburner gas temperatures and turbine performance. The resulting Stage 1 and Stage 2 cycle balances are summarized in Table 16.

Table 16 Stage 1 and Stage 2 Engine Cycle Balance

Parameter	S1 Fuel	S1 Ox.	S2 Fuel	S2 Ox.
Pump mass flow rate [kg/s]	171.5	600.3	175.2	613.1
Pump volume flow [m^3/s]	0.405	0.526	0.414	0.537
Pump inlet pressure [MPa]	0.285	0.351	0.285	0.351
Pump outlet pressure [MPa]	51.20	45.40	50.90	45.54
Pump head [m]	12259	4023	12187	4035
Pump shaft power [MW]	25.78	29.60	26.17	30.32
Preburner mixture ratio	0.25	50	0.25	50
Preburner temperature [K]	1010	858	1028	859
Preburner pressure [MPa]	40.60	41.87	41.22	42.79
Turbine mass flow rate [kg/s]	200.4	571.5	204.7	583.6
Turbine pressure ratio	1.325	1.367	1.317	1.367
Turbine outlet temperature [K]	980	806	998	807
Turbine power [MW]	28.35	32.56	28.79	33.36
Turbine power margin [%]	10.0	10.0	10.0	10.0
Injector inlet pressure [MPa]	30.64	30.64	31.30	31.30

The cycle balance provides the required 10% turbine power margin for both propellant systems on both stages. The Stage 1 fuel and oxidizer turbines provide 28.35 MW and 32.56 MW, respectively, compared with pump requirements of 25.78 MW and 29.60 MW. For Stage 2, the fuel and oxidizer turbines provide 28.79 MW and 33.36 MW compared with pump requirements of 26.17 MW and 30.32 MW.

The pressure at each station of the cycle is traced from the pump inlets to the main chamber in Table 17. The methane pump discharge passes through the cooling jacket before reaching the preburners, while the LOX pump discharge feeds both preburners directly. Both supplies must therefore exceed the highest preburner pressure by the 2.75 MPa allowance defined in Subsection II.C. The preburner gases expand through the turbines to the main injector inlet pressure of $1.20P_c$, and the 20% injector pressure drop then reduces the pressure to P_c. Pressure decreases at every station along each flow path, and any excess supply pressure is removed by the preburner control valves.

Table 17 Engine Pressure Cascade

Station [MPa]	Stage 1	Stage 2
CH_4 pump inlet (design)	0.285	0.285
CH_4 pump discharge / cooling-jacket inlet	51.20	50.90
Cooling-jacket pressure drop	−5.14	−5.36
Cooling-jacket outlet	46.06	45.54
LOX pump inlet (design)	0.351	0.351
LOX pump discharge	45.40	45.54
Required supply, $\max(P_{PB}) + 2.75$	44.62	45.54
Fuel-rich preburner	40.60	41.22
Oxidizer-rich preburner	41.87	42.79
Turbine outlet / main injector inlet, $1.20P_c$	30.64	31.30
Main injector pressure drop, $0.20P_c$	−5.11	−5.22
Chamber pressure, P_c	25.53	26.08

The heated methane from the regenerative cooling system increases the calculated preburner temperatures relative to the uncoupled cycle analysis. The Stage 1 fuel-rich and oxidizer-rich preburner temperatures are 1010 K and 858 K, respectively, compared with assumed turbine inlet temperature limits of 1000 K and 850 K. The Stage 2 temperatures are 1028 K and 859 K. These values exceed the assumed limits by approximately 1.0% and 0.9% for Stage 1 and 2.8% and 1.1% for Stage 2.

The preburner temperature is primarily controlled by the selected preburner mixture ratio. Therefore, adjustment of the fuel-rich and oxidizer-rich preburner mixture ratios is retained as an open design trade to reduce turbine inlet temperature while maintaining the required turbine power margin.

The resulting cycle balance was compared with the applicable propulsion requirements in Table 18.

Table 18 Engine Cycle Balance Requirement Assessment

Req. #	Requirement	Analysis Result	Status
1.3.1.2-08	$P_{\text{turb}} \geq 1.10 P_{\text{pump}}$	1.10 fuel / 1.10 LOX	Met
1.4.1.2-08	$P_{\text{turb}} \geq 1.10 P_{\text{pump}}$	1.10 fuel / 1.10 LOX	Met

The engine cycle therefore satisfies the turbine power requirements for both stages, with each turbine providing the required 10% power margin. The pressure balance is also sufficient to supply the regenerative cooling system, preburners, and main injector. The remaining cycle design item is refinement of the preburner mixture ratios to reduce turbine inlet temperatures below their assumed material limits while maintaining the required turbine power margin.

E. Turbopumps, Injector and Preburners

1. Turbopumps

The fuel and oxidizer turbopumps were sized using the flow rates and pressure requirements established by the engine cycle balance in Subsection II.D. The turbopumps are common to the Stage 1 and Stage 2 engine designs, with the Stage 1 operating condition used for component sizing.

The pump head is determined from the required pressure rise,

$$H = \frac{\Delta p}{\rho g_0}, \tag{9}$$

where Δp is the pump pressure rise and ρ is the propellant density. Cavitation performance was evaluated using the ratio of available to critical net positive suction head,

$$\frac{\text{NPSH}_A}{\text{NPSH}_C} \geq 1.10. \tag{10}$$

The available suction head is evaluated at the design pump inlet pressure P_{in} used in the cycle balance,

$$\text{NPSH}_A = \frac{P_{\text{in}} - P_v}{\rho g_0}, \tag{11}$$

where P_v is the propellant vapor pressure. At the 111 K methane and 90 K LOX storage temperatures, P_v = 0.0959 MPa and ρ = 422 kg/m^3 for methane, and P_v = 0.0994 MPa and ρ = 1141 kg/m^3 for LOX [23]. The inducers were sized for critical suction heads of 40.8 m (CH_4) and 17.8 m (LOX). The resulting turbopump designs are summarized in Table 19.

Table 19 Turbopump Design Summary

Parameter	Fuel (CH_4)	Oxidizer (LOX)
Volume flow rate, Q [m^3/s]	0.405	0.526
Pump head, H [m]	12259	4023
Shaft speed, N [rpm]	11162	7131
Shaft power [MW]	25.78	29.60
Design inlet pressure [MPa]	0.285	0.351
NPSH available / critical [m]	45.6 / 40.8	22.5 / 17.8
Inducer tip / hub diameter [mm]	186 / 56	236 / 71
Impeller stages	2	1
Impeller outlet diameter, D_2 [mm]	559	701
Impeller tip speed, u_2 [m/s]	327	262
Hydraulic efficiency	0.83	0.83
Cycle pump efficiency (assumed)	0.80	0.80
Turbine diameter [mm]	449	446
Turbine power [MW]	28.35	32.56

The turbopump sizing analysis calculated a hydraulic efficiency of 0.83 for both pumps. However, the engine cycle analysis assumes an overall pump efficiency of 0.80 to account for additional losses within the pump. At the design inlet

pressures, the inducers provide $\mathrm{NPSH}_A/\mathrm{NPSH}_C$ ratios of 1.12 for methane and 1.26 for LOX, exceeding the minimum requirement of 1.10. The feed system delivers higher inlet pressures than these design values (Subsection II.F), which further increases the cavitation margin in flight.

2. Injectors

The main injector was designed to deliver fuel and oxidizer into the combustion chamber through separate flow passages. Since the engine uses a full-flow staged-combustion cycle, both propellants enter the main combustion chamber as gases after passing through their respective preburners and turbines. The oxidizer-rich gas flows through the center of each injector element, while the fuel-rich gas flows through the surrounding passage. This allows the two gases to mix as they enter the combustion chamber.

The injector inlet pressure was set to $1.20P_c$, corresponding to a pressure drop equal to 20% of chamber pressure. The injector was sized using the Stage 1 operating conditions, with the resulting design summarized in Table 20.

Table 20 Main Injector Design Summary

Parameter	**Value**
Pressure drop [MPa]	5.11
Injector inlet pressure [MPa]	30.64
Oxidizer-rich gas density [kg/m^3]	146.4
Fuel-rich gas density [kg/m^3]	60.7
Number of injector elements	130
Number of injector rings	7
Center passage inner diameter [mm]	13.8
Center passage outer diameter [mm]	16.8
Outer fuel passage diameter [mm]	19.7
Oxidizer injection velocity [m/s]	200
Fuel injection velocity [m/s]	308

3. Preburners

The FFSC cycle uses separate fuel-rich and oxidizer-rich preburners to drive the fuel and oxidizer turbines. The preburner operating conditions were determined from the engine cycle balance in Subsection II.D. The preburner mixture ratios were maintained at 0.25 for the fuel-rich preburner and 50 for the oxidizer-rich preburner. The resulting Stage 1 preburner operating conditions and preliminary component dimensions are summarized in Table 21.

Table 21 Preburner Design Summary

Parameter	**Fuel-Rich**	**Oxidizer-Rich**
Pressure [MPa]	40.60	41.87
Gas temperature [K]	1010	858
Mixture ratio	0.25	50
Total flow [kg/s]	200.4	571.5
LOX flow [kg/s]	40.1	560.3
CH_4 flow [kg/s]	160.3	11.2
Chamber diameter [mm]	201	267
Chamber length [mm]	310	216
Injector elements	19	37
Injector pressure drop / preburner pressure	6.3%	6.4%

The cycle analysis predicts fuel-rich and oxidizer-rich preburner temperatures of 1010 K and 858 K, respectively, for Stage 1. These temperatures slightly exceed the assumed turbine inlet temperature limits of 1000 K and 850 K. However, all predicted temperatures for both stages remain within 5% of their respective limits. If needed, the preburner mixture ratios could be slightly adjusted to reduce the turbine inlet temperatures below these limits.

The requirements evaluated by the turbopump, injector, and preburner analysis are summarized in Table 22.

Table 22 Turbopump, Injector and Preburner Requirement Assessment

Req. #	Requirement	Analysis Result	Status
1.3.1.2-06 / 07	$\eta_{\text{pump}} \geq 0.80$	Hydraulic efficiency = 0.83; overall efficiency assumed = 0.80	Met
1.4.1.2-06 / 07	$\eta_{\text{pump}} \geq 0.80$	Hydraulic efficiency = 0.83; overall efficiency assumed = 0.80	Met
1.3.1.2-09	$\text{NPSH}_A/\text{NPSH}_C \geq 1.10$	1.26 LOX / 1.12 CH_4 at design inlet pressure	Met
1.4.1.2-09	$\text{NPSH}_A/\text{NPSH}_C \geq 1.10$	1.26 LOX / 1.12 CH_4 at design inlet pressure	Met

F. Feed System and Engine Ducts

The feed system delivers LOX and methane from their respective propellant tanks to the engine turbopump inlets. The total propellant mass flow rates are 771.86 kg/s for Stage 1 and 788.32 kg/s for Stage 2. Both engines operate at an oxidizer-to-fuel ratio of 3.5, which was used to determine the individual LOX and methane flow rates.

The analysis assumes steady, incompressible liquid flow with one circular feed duct per propellant per engine. The propellant densities were assumed constant at 1141 kg/m^3 for LOX and 422 kg/m^3 for methane. A 10% margin below the maximum allowable feed velocity of 10 m/s was applied, resulting in a design velocity of 9 m/s. Pressure losses and pipe wall thickness were neglected for the initial sizing analysis.

The minimum internal diameter of each feed duct was calculated using

$$D = \sqrt{\frac{4\dot{m}}{\pi\rho V}}, \tag{12}$$

where $\dot{m}$ is the propellant mass flow rate, ρ is the propellant density, and V is the design flow velocity. The calculated diameters were rounded upward to the next 5 mm to determine the selected internal diameters shown in Table 23.

Table 23 Selected Feed System and Engine Duct Diameters

Stage	Propellant	Mass flow (kg/s)	Minimum internal diameter (mm)	Selected internal diameter (mm)
1	LOX	600.336	272.947	275
1	Methane	171.524	239.795	240
2	LOX	613.138	275.842	280
2	Methane	175.182	242.339	245

The selected feed duct diameters were evaluated to verify that the propellant velocities remain below the maximum allowable velocity of 10 m/s. The flow velocity was calculated using

$$V = \frac{4\dot{m}}{\rho\pi D^2}, \tag{13}$$

where D is the selected internal diameter.

The pump-inlet pressure was evaluated from the tank ullage pressure, the hydrostatic head of the propellant column under the axial load factor, and the fitting losses in the feed line,

$$P_{\text{in}} = P_{\text{ull}} + \rho g_0 n \left(h_{\text{liq}} + h_{\text{line}}\right) - K\frac{\rho V^2}{2}, \tag{14}$$

where n is the axial load factor, h_{liq} is the propellant fill height above the tank outlet, h_{line} is the vertical feed-line run from the tank outlet to the pump inlet, and $K = 1.8$ is the combined minor-loss coefficient of the tank outlet, prevalve, flex joint, and elbows at the 9 m/s design velocity. Wall friction in the feed lines was neglected. The fill height was taken as $h_{\text{liq}} = H_{\text{tank}}V_{\text{prop}}/V_{\text{tank}}$, using the tank heights and volumes in Tables 3 and 4 and the propellant masses in Table 34. The LOX line run was assumed to be $h_{\text{line}} = 3.0$ m, from the tank outlet across the thrust structure to the turbopump inlet. The methane tank is located above the LOX tank, so the methane downcomer also spans the LOX tank

height, giving $h_{\mathrm{line}} = H_{\mathrm{ox}} + 3.0$ m, or 21.35 m for Stage 1 and 17.63 m for Stage 2. The intertank gap was neglected, which is conservative. Tank ullage pressures of 0.35 MPa for LOX and 0.30 MPa for methane were assumed.

The budget was evaluated at liftoff, Stage 2 ascent ignition, post-refueling ignition, and burnout of each stage. At burnout the tanks were taken as empty ($h_{\mathrm{liq}} = 0$) with a load factor of 4.0, a conservative lower bound since a lower load factor gives a lower inlet pressure. The results are summarized in Table 24.

Table 24 Pump-Inlet Pressure Budget

Condition	Propellant	Load factor	$h_{\mathrm{liq}} + h_{\mathrm{line}}$ (m)	Head (kPa)	Fitting loss (kPa)	P_{in} (MPa)	Required (MPa)
Stage 1 liftoff	LOX	1.43	20.65	330.1	83.1	0.597	0.351
Stage 1 liftoff	Methane	1.43	34.97	207.0	30.8	0.476	0.285
Stage 1 burnout	LOX	4.00	3.00	134.2	83.1	0.401	0.351
Stage 1 burnout	Methane	4.00	21.35	353.4	30.8	0.623	0.285
Stage 2 ignition	LOX	2.49	8.14	226.7	83.1	0.494	0.351
Stage 2 ignition	Methane	2.49	21.61	222.6	30.8	0.492	0.285
Post-refueling ignition	LOX	1.04	17.07	198.5	83.1	0.465	0.351
Post-refueling ignition	Methane	1.04	28.51	122.7	30.8	0.392	0.285
Stage 2 burnout	LOX	4.00	3.00	134.2	83.1	0.401	0.351
Stage 2 burnout	Methane	4.00	17.63	291.8	30.8	0.561	0.285

The minimum delivered inlet pressures are 0.401 MPa for LOX, at burnout of either stage, and 0.392 MPa for methane, at post-refueling ignition. Both exceed the design inlet pressures of 0.351 MPa (LOX) and 0.285 MPa (methane) used in the engine cycle balance (Table 16) and the turbopump design (Table 19). The turbopumps are designed to operate at these lower inlet pressures, so the additional pressure delivered by the feed system is margin. The pump pressure rise and shaft power in Table 16 are computed from the design inlet pressure and are therefore conservative, and the higher delivered pressure increases the cavitation margin. At the minimum delivered pressures, Eq. (11) gives $\mathrm{NPSH}_A/\mathrm{NPSH}_C = 1.51$ for LOX and 1.74 for methane, compared with 1.26 and 1.12 at the design inlet pressures. The resulting feed-system requirement assessment is summarized in Table 25.

Table 25 Feed System Requirement Assessment

Req. #	Requirement	Analysis Result	Status
1.3.1.2-01	$V_{\mathrm{feed,max}} \le 10$ m/s	LOX: 8.87 m/s; CH_4: 8.98 m/s	Met
1.3.1.2-02 / 03	$D_{\mathrm{LOX}} \ge 259$ mm; $D_{\mathrm{CH_4}} \ge 227$ mm	275 mm; 240 mm selected	Met
1.3.1.2-04 / 05	$P_{\mathrm{in,LOX}} \ge 0.351$ MPa; $P_{\mathrm{in,CH_4}} \ge 0.285$ MPa	0.401 MPa; 0.476 MPa minimum	Met
1.4.1.2-01	$V_{\mathrm{feed,max}} \le 10$ m/s	LOX: 8.73 m/s; CH_4: 8.81 m/s	Met
1.4.1.2-02 / 03	$D_{\mathrm{LOX}} \ge 262$ mm; $D_{\mathrm{CH_4}} \ge 230$ mm	280 mm; 245 mm selected	Met
1.4.1.2-04 / 05	$P_{\mathrm{in,LOX}} \ge 0.351$ MPa; $P_{\mathrm{in,CH_4}} \ge 0.285$ MPa	0.401 MPa; 0.392 MPa minimum	Met

G. Thrust Vector Control

A thrust vector control (TVC) system is required to provide pitch and yaw control during powered flight. Four TVC concepts were evaluated for the COPE vehicle: electromechanical engine gimbaling, hydraulic engine gimbaling, differential throttling, and jet vanes. The concepts were compared based on control authority, reliability, mass, specific impulse penalty, support-system requirements, heritage, and packaging.

Table 26 Thrust Vector Control Trade Study

Criterion	Weight	Gimbal + EMA	Gimbal + Hydraulic	Differential Throttle	Jet Vanes
Control authority	0.20	5	5	2	3
Reliability and simplicity	0.20	4	3	3	2
Mass	0.15	4	3	5	3
Specific impulse penalty	0.15	5	5	4	1
Power and fluids	0.10	3	4	5	5
Heritage at high thrust	0.10	4	5	2	2
Packaging	0.10	4	3	5	3
Weighted Total	**1.00**	**4.25**	**4.00**	**3.55**	**2.60**

The electromechanical actuator (EMA) gimbal system received the highest weighted score and was selected for both stages. Engine gimbaling provides direct thrust-vector control without the specific impulse losses associated with jet vanes. Additionally, electromechanical actuators eliminate the hydraulic pumps, fluid lines, and reservoirs required by a hydraulic gimbal system.

The required gimbal authority was evaluated at the maximum dynamic pressure condition, where aerodynamic disturbances are greatest. The analysis used the 95% Atmospheric Model Reference wind condition [24] with no aerodynamic load relief, an angle of attack of approximately 11°, a normal-force slope of $C_{N_\alpha} = 3.0\ \mathrm{rad}^{-1}$, and an assumed engine misalignment of 0.25°. A factor of 1.5 was applied to the calculated disturbance moment to provide additional control authority.

The required gimbal angle was determined by balancing the TVC moment against the aerodynamic and engine-misalignment disturbance moments,

$$T(x_{cg} - x_g)\sin\delta \geq 1.5\,(M_{\mathrm{aero}} + M_{\mathrm{mis}})\,, \tag{15}$$

where T is the stage thrust, $x_{cg} - x_g$ is the distance between the vehicle center of gravity and the engine gimbal point, δ is the engine gimbal angle, and M_{aero} and M_{mis} are the aerodynamic and engine-misalignment disturbance moments, respectively.

The actuator force was determined from the required gimbal torque. A preliminary force margin of 25% was included in the actuator sizing,

$$F_{\mathrm{act}} = 1.25\frac{\tau_{\mathrm{gimbal}}}{r_a}, \tag{16}$$

where τ_{gimbal} is the required gimbal torque and r_a is the actuator moment arm. Using the calculated gimbal torques of 77.0 kN · m and 123.1 kN · m with a moment arm of 0.600 m gives

$$F_{\mathrm{act},1} = 1.25\left(\frac{77.0}{0.600}\right) = 160.4\ \mathrm{kN}, \tag{17}$$

$$F_{\mathrm{act},2} = 1.25\left(\frac{123.1}{0.600}\right) = 256.5\ \mathrm{kN}. \tag{18}$$

The Stage 1 and Stage 2 TVC sizing results are summarized in Table 27.

Table 27 Thrust Vector Control and Actuator Sizing

Parameter	Stage 1	Stage 2
Required nominal gimbal angle [°]	5.05	–
Engine-out gimbal angle [°]	6.47	3.89
Gimbal torque [kN·m]	77.0	123.1
Actuator force [kN]	160.4	256.5
Actuator stroke [mm]	±63	±63
Actuator rate [m/s]	0.105	0.105
Electrical power per actuator [kW]	22.4	35.8
Total TVC peak power [kW]	358	215
Gimbal pin diameter [mm]	65	70
Actuator moment arm [mm]	600	600

The nominal Stage 1 gimbal requirement of 5.05° remains within the selected ±6° mechanical gimbal range. Since the required angle already includes the 1.5 disturbance-moment factor, the nominal configuration provides sufficient control authority under the analyzed conditions. The available TVC moment at 6° is approximately 19% greater than the moment required at 5.05°, assuming constant thrust and moment arm.

A single-engine-out condition increases the required Stage 1 gimbal angle to 6.47°, exceeding the mechanical limit by 0.47°. Differential throttling of the opposite outer engine is proposed to provide the additional control moment while maintaining the ±6° mechanical gimbal range. The Stage 2 engine-out condition requires 3.89° of gimbal authority at trans-lunar injection, which remains within the selected mechanical range.

The applied actuator loads of 128.3 kN for Stage 1 and 205.2 kN for Stage 2 remain below the actuator design load ratings of 160.4 kN and 256.5 kN, giving a margin of safety of $MS = 160.4/128.3 - 1 = 0.25$ for each stage. Final actuator selection must confirm that the selected actuators provide these rated loads, stroke lengths, and operating rates.

The total peak TVC power requirements are approximately 358 kW for Stage 1 and 215 kW for Stage 2. Because these transient power demands exceed the primary avionics power allocation, dedicated high-power battery packs are proposed to supply the TVC system.

The TVC requirements evaluated by this analysis are summarized in Table 28.

Table 28 Thrust Vector Control Requirement Assessment

Req. #	Requirement	Analysis Result	Status
1.3.1.3-01	$\lvert\delta_{\max}\rvert \geq 6°$	5.05° required; ±6° available	Met
1.3.1.3-02	$M_{\text{TVC}} \geq 1.5 M_{\text{disturbance}}$	1.5 disturbance moment factor applied	Met
1.3.1.3-03	$F_{\text{act,applied}} \leq 160.4$ kN	128.3 kN	Met
1.4.1.3-01	$\lvert\delta_{\max}\rvert \geq 6°$	3.89° required; ±6° available	Met
1.4.1.3-02	$M_{\text{TVC}} \geq 1.5 M_{\text{disturbance}}$	1.5 disturbance moment factor applied	Met
1.4.1.3-03	$F_{\text{act,applied}} \leq 256.5$ kN	205.2 kN	Met

H. Structures

The Stage 1 and Stage 2 propellant tanks were sized based on the required propellant mass, propellant density, and a 95% fill fraction. The current design uses LOX and CH_4 at an O/F ratio of 3.5 with a common vehicle diameter of 6.9 m. LOX and CH_4 densities of 1141 kg/m^3 and 422 kg/m^3 were used to determine the required tank volumes and heights. The resulting tank dimensions are summarized in Table 29.

Table 29 Propellant Tank Sizing

Parameter	S1 LOX	S1 CH_4	S2 LOX	S2 CH_4
Propellant Mass [kg]	721416	206119	616313	176090
Tank Volume [m^3]	665.54	514.14	568.58	439.24
Tank Height [m]	17.80	13.75	15.21	11.75

The refueling vehicle uses the Stage 2 tanks without modification. Its 252141 kg LOX and 72040 kg CH_4 launch load (Subsection II.I) fills 40.9% of the Stage 2 tank capacity, so no separate refueling-vehicle tank sizing is required.

The interstage must provide enough length for the Stage 2 thrust chamber and clearance between the two stages. Using the 4.08 m Stage 2 thrust-chamber length from Table 9 and a 0.50 m clearance gives

$$L_{\text{interstage}} \geq L_{\text{TC,S2}} + \Delta_{\text{clear}} = 4.08 + 0.50 = 4.58 \text{ m}. \tag{19}$$

A 4.70 m interstage was selected, providing 0.12 m of additional clearance.

The overall vehicle height was determined by combining the major Stage 1, Stage 2, interstage, and crew capsule dimensions. The tank stacks include the cylindrical tank sections and common-bulkhead dome geometry. The resulting vehicle stack-up is shown in Table 30.

Table 30 Vehicle Height Stack-Up

Vehicle Segment	Length [m]
Stage 1 engine and thrust structure	6.43
Stage 1 tank stack	32.70
Stage 1-2 interstage	4.70
Stage 2 aft section	2.57
Stage 2 tank stack	28.11
Orion crew and launch abort system [26]	20.42
Total Vehicle Height	**94.93**

The resulting vehicle height is 94.93 m, satisfying the 95 m overall vehicle height requirement while maintaining the required Stage 1 and Stage 2 propellant tank volumes. The vehicle is shorter than the 98 m SLS Block 1 already processed and launched at LC-39B [27].

The primary structural loads were evaluated at liftoff, immediately before Stage 1 shutdown, and following Stage 1 separation. The total thrust, resulting vehicle acceleration, and compressive load transmitted through each structural interface are calculated using

$$F_T = N_{\text{eng}} F_{\text{eng}}, \qquad a_T = \frac{F_T}{M}, \qquad F_{\text{interface}} = m_{\text{supported}} a_T \tag{20}$$

The Stage 1-2 interstage supports the complete upper-stage assembly, while the capsule adapter supports the crew capsule and payload. The resulting axial loads for each flight condition are shown in Table 31.

Table 31 Vehicle Axial Load Conditions

Parameter	Liftoff	Stage 1 Vacuum	Stage 2 Vacuum
Vehicle Mass [kg]	1432341	504806	356699
Total Thrust [MN]	20.056	21.840	8.715
Acceleration [m/s^2]	14.00	43.26	24.43
Interstage Load [MN]	4.995	15.432	–
Capsule Adapter Load [MN]	0.441	1.364	0.770

The Stage 1 vacuum condition produces the maximum structural loading due to the reduced vehicle mass near the end of the Stage 1 burn. The maximum Stage 1-2 interstage load is 15.432 MN, remaining below the 15.5 MN design load requirement. The maximum capsule adapter load is 1.364 MN, remaining below the 1.4 MN design load requirement.

The Stage 1-2 interstage was modeled as a cylindrical shell with a 6.9 m outside diameter and a preliminary wall thickness of 4.0 mm. The capsule adapter was modeled as a conical shell, with the axial stress corrected for the adapter angle. The resulting stresses for each flight condition are shown in Table 32.

Table 32 Calculated Structural Stresses

Structural Component	Liftoff	Stage 1 Vacuum	Stage 2 Vacuum
Interstage Axial Stress [MPa]	57.64	178.08	–
Capsule Adapter Axial Stress [MPa]	31.20	96.40	54.44
Capsule Adapter Meridional Stress [MPa]	31.52	97.40	55.01

The maximum interstage axial stress is 178.08 MPa and the maximum capsule adapter meridional stress is 97.40 MPa, both occurring during the Stage 1 vacuum condition. These values remain below the design limits of 180 MPa and 100 MPa, respectively. The structural requirements evaluated by this analysis are summarized in Table 33.

Table 33 Structures Requirement Assessment

Req. #	Requirement	Analysis Result	Status
1-03	$D = 6.9$ m	6.9 m design diameter	Met
1-04	$H \leq 95$ m	94.93 m	Met
1.3.2-02	$H_{ox} \leq 18.35$ m	17.80 m	Met
1.3.2-03	$H_{fuel} \leq 14.17$ m	13.75 m	Met
1.3.2.1-01	$V_{LOX} \geq 665$ m^3	665.54 m^3	Met
1.3.2.2-01	$V_{CH_4} \geq 514$ m^3	514.14 m^3	Met
1.3.2.3-01	$F_{\text{axial,max}} \leq 15.5$ MN	15.432 MN	Met
1.3.2.3-02	$\sigma_{\text{axial,max}} \leq 180$ MPa	178.08 MPa	Met
1.3.2.3-03	$L_{\text{interstage}} \geq 4.7$ m	4.70 m selected	Met
1.4.2-02	$H_{ox} \leq 15.25$ m	15.21 m	Met
1.4.2-03	$H_{fuel} \leq 11.78$ m	11.75 m	Met
1.4.2.1-01	$V_{LOX} \geq 568$ m^3	568.58 m^3	Met
1.4.2.2-01	$V_{CH_4} \geq 439$ m^3	439.24 m^3	Met
1.4.2.3-01	$F_{\text{axial,max}} \leq 1.4$ MN	1.364 MN	Met
1.4.2.3-02	$\sigma_{\text{meridional,max}} \leq 100$ MPa	97.40 MPa	Met
1.5.1-02 / 03	$M_{ox} \leq 252141$ kg; $M_f \leq 72040$ kg	252141; 72040 kg in the S2 tanks (40.9% full)	Met
1.5.2-02 / 03	$H_{ox} \leq 18.35$ m; $H_{fuel} \leq 14.17$ m	15.21 m; 11.75 m (S2 tanks)	Met

I. Mass Properties and Inertia Matrix

A mass-properties model was developed to determine the vehicle mass, propulsion-system center of gravity (CG), and moments of inertia. Individual components were modeled using their estimated masses, geometries, and locations within the vehicle. Mass growth allowances were applied to account for uncertainties in the preliminary component designs.

The vehicle mass model is summarized in Table 34. The Stage 1 ascent configuration includes the upper-stage vehicle, while Stage 2 is evaluated before and after orbital refueling for trans-lunar injection.

Table 34 Vehicle Mass Model

Parameter	Stage 1 Ascent	Stage 2 Ascent	Stage 2 Post-Refuel
Initial mass, M_0 [kg]	1432341	356699	856441
Propellant mass, M_p [kg]	927535	292661	792403
Structure mass, M_s [kg]	148107	32518	32518
Stage thrust [MN]	20.05	8.71	8.71
Initial thrust-to-weight	1.43	2.49	1.04
Burn time [s]	150	124	335
ΔV [m/s]	3622	6328	9555

The Stage 1 structural mass is 148107 kg, which is 3.8% below its 154000 kg allocation. The Stage 2 structural mass is 32518 kg, approximately 0.06% above its 32500 kg allocation and within the 5% design tolerance. The Stage 1 mass margin offsets the small Stage 2 overage while maintaining the overall vehicle mass below its allocated value. The Stage 1 ascent mass includes the allocated 5000 kg mission payload and the complete upper-stage vehicle.

Component masses were estimated using the current best estimate (CBE) and a mass growth allowance (MGA). The resulting mass estimate value (MEV) was calculated using

$$m_{\text{MEV}} = m_{\text{CBE}}(1 + \text{MGA}), \tag{21}$$

where MGA is expressed as a fraction of the estimated component mass. Allowances ranging from 5% to 20% were applied based on component maturity and design uncertainty. The engine mass breakdown is summarized in Table 35.

Table 35 Engine Component Mass Breakdown

Component	S1 CBE [kg]	MGA	S1 MEV [kg]	S2 MEV [kg]
Chamber liner	68	10%	74	138
Closeout jacket	219	10%	241	537
Coolant and hot-gas manifolds	29	20%	34	74
Main injector	215	20%	258	264
Fuel turbopump	522	20%	626	626
LOX turbopump	600	20%	720	720
Preburners (2)	160	20%	192	192
Hot-gas ducts / turbine housings	90	20%	108	108
Main and preburner valves	150	15%	172	172
Engine propellant ducts / flex joints	60	15%	69	69
Gimbal block and thrust mount	75	15%	86	100
Torch igniters and exciters	12	5%	13	13
Controller, harness, and sensors	45	5%	47	47
Base heat shield	40	15%	46	46
Engine Total	**2284**	–	**2688**	**3106**
Mass Allocation	–	–	**2800**	**3300**

The Stage 1 engine has an MEV of 2688 kg, approximately 4% below its 2800 kg allocation. The Stage 2 engine has an MEV of 3106 kg, approximately 6% below its 3300 kg allocation. Both engines remain within their allocated mass limits after applying the mass growth allowances.

The engine masses were combined with the feed systems, pressurization hardware, TVC components, and associated propulsion equipment to estimate the total dry propulsion-system mass. Stage 1 contains eight engines, while Stage 2 contains three engines. The resulting mass breakdown is provided in Table 36.

Table 36 Propulsion-System Mass Rollup

Item	S1 CBE [kg]	S1 MGA	S1 MEV [kg]	S2 CBE [kg]	S2 MGA	S2 MEV [kg]
Main engines	18274	17.7%	21501	7982	16.7%	9318
LOX manifold and branches	877	20%	1052	256	20%	307
CH_4 downcomer	1579	20%	1895	757	20%	909
CH_4 branches and flex joints	651	20%	782	246	20%	295
Prevalves	640	15%	736	240	15%	276
Autogenous heat exchangers	240	20%	288	90	20%	108
Pressurant lines, regulators, and vents	488	20%	586	404	20%	485
TVC actuators	1120	20%	1344	420	20%	504
TVC power and control	798	15%	918	449	15%	516
Total Propulsion Dry Mass	**24668**	**18.0%**	**29102**	**10843**	**17.3%**	**12717**

The main-engine MGA is the mass-weighted value of the component allowances in Table 35. The remaining allowances follow the same maturity-based categories of AIAA S-120A [14]: 20% for items sized from mass-estimating relationships and 15% for modified existing hardware such as prevalves and TVC power units. Including these allowances, the dry propulsion-system mass grows from 35511 kg (CBE) to 41819 kg (MEV), a total margin of 6308 kg (17.8%).

The propulsion-system CG and inertia were calculated from the mass and location of each component. Components were approximated as cylinders, disks, rings, or thin shells based on their geometry. The Stage 1 propulsion configuration consists of one central engine and seven surrounding engines positioned on a 2.35 m radius ring. Stage 2 uses three engines arranged on a 1.85 m radius ring.

The vehicle X-axis is defined forward from the Stage 1 engine exit plane. The center of gravity was calculated using

$$\mathbf{r}_{CG} = \frac{\sum_i m_i \mathbf{r}_i}{\sum_i m_i}, \tag{22}$$

where m_i is the component mass and $\mathbf{r}_i$ is its position relative to the selected reference point.

The inertia tensor was calculated by combining the individual component inertias using the parallel-axis theorem,

$$\mathbf{I}_{CG} = \sum_i \left[\mathbf{R}_i \mathbf{I}_i \mathbf{R}_i^T + m_i \left(\|\mathbf{d}_i\|^2 \mathbf{1} - \mathbf{d}_i \mathbf{d}_i^T \right) \right], \tag{23}$$

where $\mathbf{I}_i$ is the component inertia tensor about its own CG, $\mathbf{R}_i$ accounts for component orientation, and $\mathbf{d}_i$ is the component position relative to the system CG.

The calculated mass properties are summarized in Table 37.

Table 37 Propulsion-System Mass Properties

System	MEV [kg]	X_{CG} [m]	$I_{xx}/I_{yy}/I_{zz}$ [kg·m^2]
Stage 1 engine	2688	1.366	1075 / 1790 / 2069
Stage 2 engine	3106	1.706	1713 / 4930 / 5228
Stage 1 propulsion, dry	29102	3.23	152347 / 799027 / 801186
Stage 2 propulsion, dry	12717	46.32	51013 / 322852 / 323717
Entire propulsion system, dry	41819	16.34	203362 / 17553888 / 17556911

All masses, CG locations, and inertias in Table 37 and Eqs. (24)–(26) are computed with MEV masses, so the inertia matrices include the mass growth allowance. The combined dry propulsion system has an estimated mass of 41819 kg and a center of gravity located 16.34 m forward of the Stage 1 engine exit plane. Stage 1 accounts for approximately 70% of the total dry propulsion mass, while Stage 2 accounts for the remaining 30%. The transverse moments of inertia are substantially larger than the axial moment because of the longitudinal separation between the two stages.

The calculated Stage 1 propulsion-system inertia matrix is

$$\mathbf{I}_{S1} = \begin{bmatrix} 152347 & -3654 & -7104 \\ -3654 & 799027 & 1758 \\ -7104 & 1758 & 801186 \end{bmatrix} \text{ kg} \cdot \text{m}^2. \tag{24}$$

The calculated Stage 2 propulsion-system inertia matrix is

$$\mathbf{I}_{S2} = \begin{bmatrix} 51013 & -1253 & -2547 \\ -1253 & 322852 & 709 \\ -2547 & 709 & 323717 \end{bmatrix} \text{ kg} \cdot \text{m}^2. \tag{25}$$

The combined dry propulsion-system inertia matrix is

$$\mathbf{I}_{\text{prop}} = \begin{bmatrix} 203362 & -8232 & -14010 \\ -8232 & 17553888 & 2466 \\ -14010 & 2466 & 17556911 \end{bmatrix} \text{ kg} \cdot \text{m}^2. \tag{26}$$

The combined inertia tensor includes the contribution of each stage about the overall propulsion-system CG. The off-diagonal terms account for the current three-dimensional component locations and are retained in the combined inertia matrix rather than assuming a perfectly symmetric propulsion system.

The vehicle mass and velocity results were compared with the applicable system requirements. The Stage 1 ascent mass is 1432341 kg, remaining below the 1460000 kg vehicle mass allocation. The 5000 kg mission payload is included in the ascent mass model. The Stage 1 structural mass is 148107 kg, or 3.8% below its 154000 kg allocation. The Stage 2 structural mass is 32518 kg, approximately 0.06% above its 32500 kg allocation and within the 5% design tolerance.

The Stage 1 trajectory provides 3622 m/s of ΔV, while Stage 2 provides 6328 m/s during ascent and 9555 m/s following orbital refueling. The total mission capability is therefore 19505 m/s, satisfying the 19.5 km/s vehicle requirement. The applicable mass and velocity requirements are summarized in Table 38.

The refueling vehicle uses the same Stage 2 vehicle architecture, including the propulsion system, primary structure, and propellant tanks. It launches at the same 356699 kg initial mass as the crewed Stage 2, with the 26520 kg crew capsule and 5000 kg payload replaced by 31520 kg of additional propellant. The refueling vehicle propellant load is therefore $M_p = 292661 + 31520 = 324181$ kg, or 252141 kg of LOX and 72040 kg of CH_4 at $O/F = 3.5$. Because the initial mass, structural mass, and engines match the crewed Stage 2, the Stage 2 propulsion and structural analyses apply to the refueling vehicle.

Table 38 Mass Properties Requirement Assessment

Req. #	Requirement	Analysis Result	Status
1-01	$\Delta V = 19.5$ km/s	19.505 km/s total	Met
1-02 / 1.3-03	$M_0 \leq 1460000$ kg	1432341 kg	Met
1.3-02	$M_i \leq 820000$ kg	$M_b = 504806$ kg	Met
1.3.1-02 / 03	$M_{ox} \leq 737000$; $M_f \leq 211000$ kg	721416; 206119 kg (−2.1%; −2.3%)	Met
1.4-02	$M_i \leq 82000$ kg	$M_b = 64038$ kg	Met
1.4-03 / 1.5-01	$M_0 = 356699$ kg	356699 kg (S2 and refueling vehicle)	Met
1.4.1-02 / 03	$M_{ox} = 228000$; $M_{pr} = 65000$ kg	227625; 65036 kg	Met
1.4.1-05 / 06	$M_{ox} = 588000$; $M_{pr} = 168000$ kg	616313; 176090 kg (+4.8%)	Met (Within 5%)
1.1-01	$M_s = 5000$ kg	5000 kg payload included in mass model	Met
1.3-01	$\Delta V = 3650$ m/s	3622 m/s (−0.8%)	Met (Within 5%)
1.3.1-01	$\Delta V = 3.6$ km/s	3.622 km/s	Met
1.3.2-01	$M_s \leq 154000$ kg	148107 kg (−3.8%)	Met
1.4.1-01 / 1.5.1-01	$\Delta V = 3.6$ km/s	6.328 km/s available	Met
1.4.1-04	$\Delta V = 9.5$ km/s	9.555 km/s	Met
1.4.2-01	$M_s \leq 32500$ kg	32518 kg (+0.06%)	Met (Within 5%)
1.5.2-01	$M_s \leq 32500$ kg	32518 kg (+0.06%)	Met (Within 5%)
1.3.1.1-09	$m_{\text{eng}} \leq 2800$ kg	2688 kg MEV	Met
1.4.1.1-08	$m_{\text{eng}} \leq 3300$ kg	3106 kg MEV	Met

J. COTS and Technology Readiness

Commercially available and heritage hardware were incorporated into the COPE design where practical to reduce development risk and avoid unnecessary redesign of established aerospace systems. The vehicle uses existing technologies for crew support, avionics, guidance, thermal control, propulsion components, stage separation, and ground operations.

Technology readiness levels (TRLs) were evaluated using NASA's nine level classification system [16]. TRL 9 represents technology proven through successful mission operations, while TRLs 6 and 7 require prototype demonstrations in relevant and operational environments, respectively. Since COPE adapts existing technologies to a new vehicle configuration, the maturity of the reference technology is distinguished from the maturity of its COPE-specific implementation.

Table 39 summarizes the commercial and heritage hardware identified for the vehicle. The listed TRLs are preliminary assessments of the reference technologies and do not establish qualification of the corresponding COPE systems. Components identified as hardware classes require final sizing, selection, and verification against the applicable operating conditions. Specific suppliers, model numbers, and source information are provided in Appendix A.3.

Table 39 Commercial and Heritage Hardware Identified for COPE

WBS	Hardware / Reference Technology	Applicable Capability	TRL
Crew Capsule and Payload Systems			
1.2 / 1.2.1 / 1.2.7-03 / 1.2.7-04	Lockheed Martin / Collins Aerospace: Orion crew module and ECLSS class	Crew-support heritage for four astronauts and missions exceeding the COPE requirement of three astronauts for 14 days.	9
1.2.2	Lockheed Martin / Collins Aerospace: Orion active thermal-control system and Avcoat TPS	Heritage cabin temperature regulation and lunar-return thermal protection.	9
1.2.3 / 1.2.7-01 / 1.2.7-02	Northrop Grumman: Orion Launch Abort System abort motor	Approximately 1780 kN abort thrust compared with the 1040 kN COPE requirement.	9

Table 39 (Continued)

WBS	Hardware / Reference Technology	Applicable Capability	TRL
1.2.5	Airbus / ESA: Orion European Service Module solar arrays and batteries	Heritage spacecraft power generation and storage technology applicable to the 28 VDC architecture.	9
1.2.6	Aerojet Rocketdyne: MR-107 and MR-104 monopropellant thruster families	Approximately 110 N attitude-control thrust, with higher-thrust options for larger maneuvers.	9
1.1.1	Northrop Grumman: High Efficiency Cryocooler class	Pulse-tube cryogenic cooling technology applicable to the 40 K payload-storage requirement.	8
Propulsion and Thrust Vector Control			
1.3.1.3 / 1.4.1.3	Moog: Modular Electromechanical Actuator (MEMA) family	Reference actuator technology for Stage 1 and Stage 2 TVC forces of approximately 160 and 256 kN, respectively.	8
1.3.1.3 / 1.4.1.3	Moog: TVC actuation controller	Reference controller technology for redundant electromechanical actuator control and engine gimbal positioning.	8
1.3.1.1 / 1.4.1.1	Kulite: XTEH-10L-190 and HKM-375 pressure transducers	High-temperature and cryogenic pressure measurement technology for engine chambers, tanks, and feed systems.	8
1.3.1.2 / 1.4.1.2	Marotta Controls: CoRe MV100X and MV524X cryogenic valves and space-qualified pressure regulators	Cryogenic valve and regulator technology applicable to LOX, CH_4, and tank-pressurization systems.	8
1.3.1.2 / 1.4.1.2	Valcor Engineering: Cryogenic launch vehicle valves	Reference ball and butterfly valve technology for 275-280 mm LOX and 240-245 mm CH_4 feed lines.	8
1.3.1.1 / 1.4.1.1	Unison Industries: Solid-state ignition exciters and augmented spark igniters	Reference ignition technology for the main combustion chamber and preburners, including repeated engine starts.	8
1.3.1.2 / 1.4.1.2	Senior Aerospace Bird Bellows: Flexible feed-line joints	Gimbal-compensating bellows technology applicable to the required $\pm 6^\circ$ engine motion.	8
1.3.1.1 / 1.4.1.1	Aerojet Rocketdyne / Moog: High-pressure cryogenic engine valves	Heritage main and preburner valve technology for high-pressure LOX and CH_4 flow paths.	6
1.3.1.2 / 1.4.1.2	Timken / Barden: Hybrid angular-contact bearings	Hybrid ceramic bearing technology applicable to high-speed cryogenic turbopump operation.	8
1.3.1.2 / 1.4.1.2	The Lee Company: HI-BAR safety screens	Contamination protection for auxiliary fluid and pressurization lines.	8
1.3.1.3 / 1.4.1.3	Moog: Reconfigurable Integrated Battery (RIB), High Power	Dedicated high-power battery architecture for TVC operation, separate from the spacecraft avionics bus.	6
Stage Separation and Safety Systems			
1.3.5	Ensign-Bickford Aerospace & Defense: Frangible joint assemblies	Heritage stage-separation technology applicable to rapid separation with controlled debris generation.	9
1.3.5	Ensign-Bickford Aerospace & Defense: Separation bolts and pusher systems	Reference separation hardware for achieving the required 1.5 m/s stage-separation velocity.	9
1.3.5 / 1.2.3	Pacific Scientific Energetic Materials: NASA Standard Initiator class	Heritage pyrotechnic initiation technology applicable to redundant separation and abort systems.	9
1.3.6 / 1.4.6 / 1.5.6	Aerojet Rocketdyne: MR-107 monopropellant thruster	Approximately 110 N reaction-control thrust applicable to vehicle attitude and roll control.	9

Table 39 (Continued)

WBS	Hardware / Reference Technology	Applicable Capability	TRL
Thermal Control, Electrical, and Avionics			
1.3.3 / 1.4.3 / 1.5.3	Dunmore: Double-Aluminized Kapton multilayer insulation	Heritage cryogenic insulation technology applicable to LOX and CH_4 tank thermal protection.	8
1.1.1 / 1.3.3 / 1.4.3 / 1.5.3	Lake Shore Cryotronics: DT-670-SD silicon diode temperature sensor	Cryogenic temperature measurement for the 40 K payload storage and the 90 K LOX and 111 K CH_4 tanks.	8
1.2.2 / 1.3.3 / 1.4.3 / 1.5.3	Minco: Polyimide flexible survival heaters	Heritage survival heating for avionics and component thermal control.	9
1.3.1.3 / 1.4.4 / 1.5.4	Saft: VL51ES lithium-ion battery cells	Reference high-rate lithium-ion technology for spacecraft electrical-power storage and distribution.	8
1.3.7 / 1.4.7 / 1.5.7 / 1.2.7	Moog: Integrated Avionics Unit and Main Avionics Controller	Heritage avionics technology supporting redundant control, fault detection, and vehicle safety functions.	8
1.2.6 / 1.4.6 / 1.5.6	Honeywell: Miniature Inertial Measurement Unit (MIMU)	Heritage inertial attitude and rate measurement for guidance, navigation, and control.	9
1.2.6 / 1.4.6 / 1.5.6	Sodern: Hydra star tracker	Heritage absolute attitude determination for in-space navigation and docking.	9
Refueling, Docking, and Ground Systems			
1.4.5 / 1.5.5	Ship-to-ship cryogenic quick-disconnect system	Development hardware for 100 kg/s LOX/CH_4 transfer and repeated orbital mating.	5
1.6	NASA Kennedy Space Center: LC-39B launch facility	Existing heavy-lift launch infrastructure applicable to vertical launch and ground-support operations.	9
1.5.6 / 1.4.5	NASA / Boeing: International Docking System Standard class	Heritage docking technology providing a reference for autonomous alignment, soft capture, and spacecraft mating.	9

The hardware identified in Table 39 demonstrates that many of the capabilities required by COPE are supported by established aerospace technologies. Flight-proven components and heritage designs provide a foundation for the vehicle's propulsion, structural, electrical, and control systems. However, the maturity of an existing technology does not automatically establish its suitability for the COPE operating environment.

Table 40 compares the demonstrated maturity of the reference technologies with the preliminary maturity of the COPE-specific designs. The reference TRLs represent existing technology heritage, while the COPE TRLs reflect the current development status. Since COPE-specific hardware has not yet undergone physical testing, the assigned design TRLs are preliminary and do not represent completed qualification.

Table 40 Technology Readiness Assessment for COPE-Designed Hardware

WBS	Technology	Reference TRL	COPE TRL	Remaining Verification and Development
1.3.1.1 / 1.4.1.1	LOX/CH_4 full-flow staged-combustion engine	9	3	Heritage-based engine design requiring component, powerpack, and integrated hot-fire testing at the selected operating conditions.
1.3.1.1	Oxidizer-rich preburner and turbine	9	3	Existing turbine technology provides a design reference. Material compatibility, combustion performance, and turbine operation require testing.
1.3.1.1	Fuel-rich CH_4 preburner and turbine	8-9	3	Heritage turbine technology supports the design. Hot-fire testing must verify combustion performance, thermal limits, and carbon deposition.
1.3.1.1	GRCop-42 regenerative chamber liner	6	3	Existing material and manufacturing demonstrations support the design. The COPE chamber requires fabrication, pressure testing, and hot-fire verification.
1.3.1.1	NASA HR-1 closeout jacket	6	3	Established material and manufacturing research supports the design. Fabrication, bond inspection, and structural and thermal testing remain necessary.
1.3.1.2 / 1.4.1.2	Autogenous GOX/GCH_4 pressurization	9	3	Existing pressurization technology provides a design basis. Integrated testing must verify heat-exchanger performance, pressure regulation, and tank response.
1.3.1.3 / 1.4.1.3	Electromechanical TVC system	9	3	Existing actuator and control technology supports the design. Integrated testing must verify actuator forces, gimbal motion, control response, and power demand.
1.3.2 / 1.4.2	Al-Li 2195 welded cryogenic tanks	9	3	Flight-proven materials and manufacturing methods provide a design basis. Representative COPE tanks require weld inspection, structural testing, and cryogenic qualification.
1.4.5 / 1.5.5	Ship-to-ship cryogenic propellant transfer	4-5	2-3	Cryogenic transfer demonstrations provide a technology basis. High-rate LOX/CH_4 transfer, repeated orbital mating, and interface sealing require testing.
1.4.5 / 1.5.5	Long-duration LOX/CH_4 storage	5	3	Existing cryogenic storage technology supports the design. Thermal-vacuum testing must verify insulation performance, boil-off rates, and propellant management.

The assessment indicates that COPE relies primarily on established aerospace technologies rather than requiring the development of new operating principles. The reference technologies provide a strong foundation for the vehicle design and reduce development risk. However, the COPE-specific configurations require additional testing before higher technology readiness levels can be demonstrated.

The primary development risks are associated with the integrated propulsion system, orbital cryogenic propellant transfer, and long-duration cryogenic storage. These systems require component testing, integrated demonstrations, and qualification under representative mission conditions. Successful completion of these activities will support advancement toward flight-ready maturity.

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Appendix

A. CFD Analysis

The following figures provide the complete plume contour results from the Stage 1 and Stage 2 nozzle CFD analyses. Temperature, pressure, and Mach number contours are included to show the exhaust plume expansion and near-field flow behavior downstream of each nozzle.

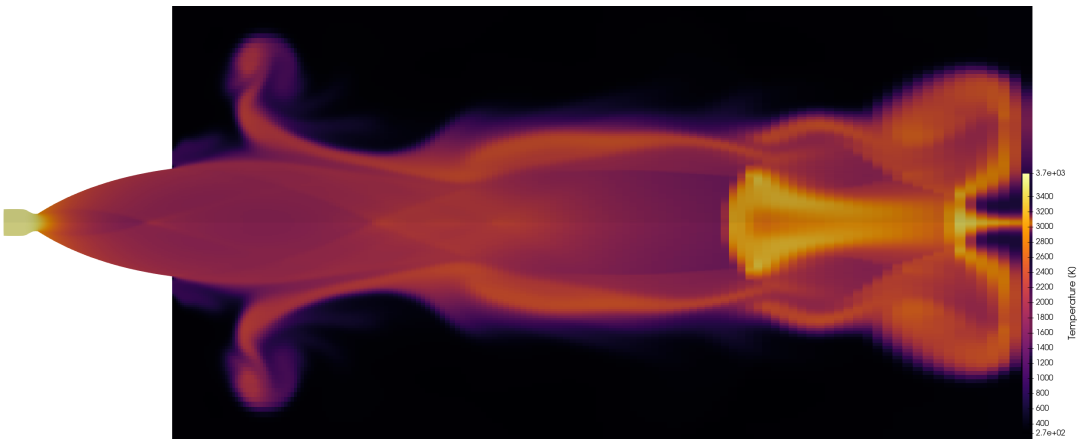


Fig. 1 Stage 1 exhaust plume temperature contours.

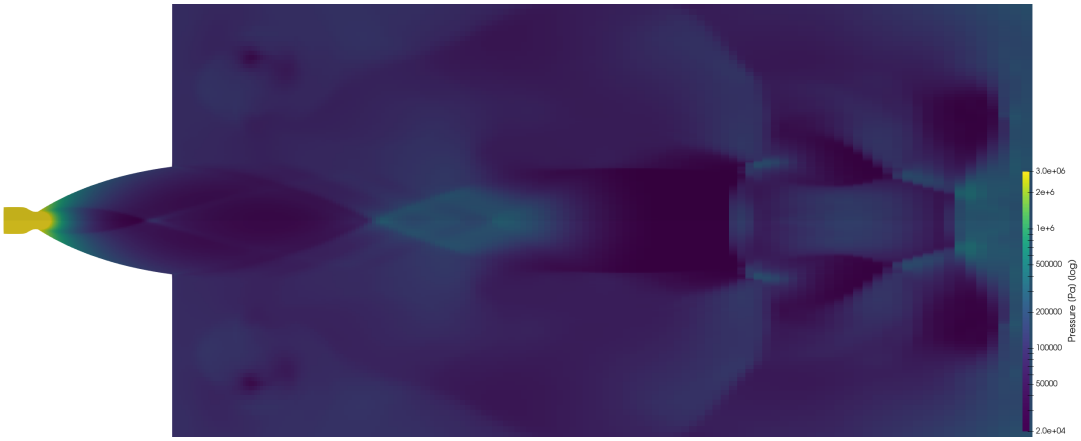


Fig. 2 Stage 1 exhaust plume pressure contours.

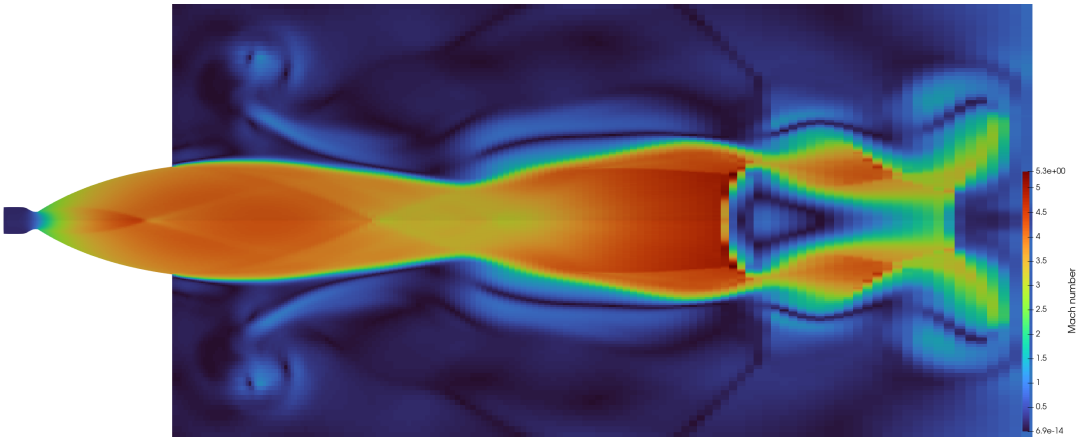


Fig. 3 Stage 1 exhaust plume Mach number contours.

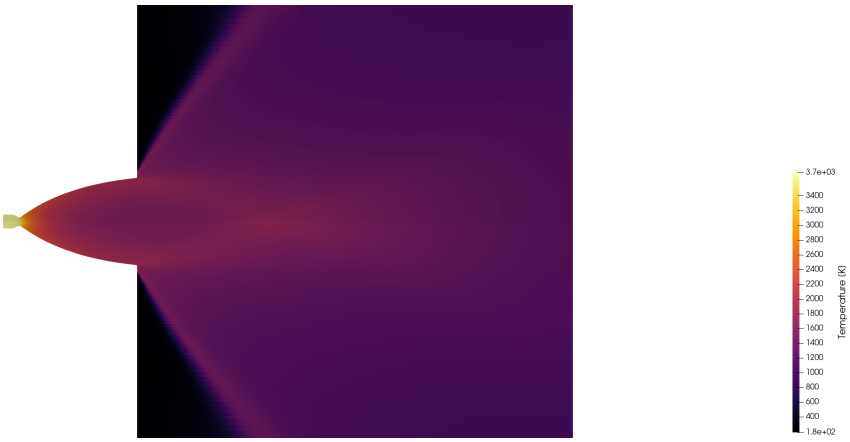


Fig. 4 Stage 2 exhaust plume temperature contours.

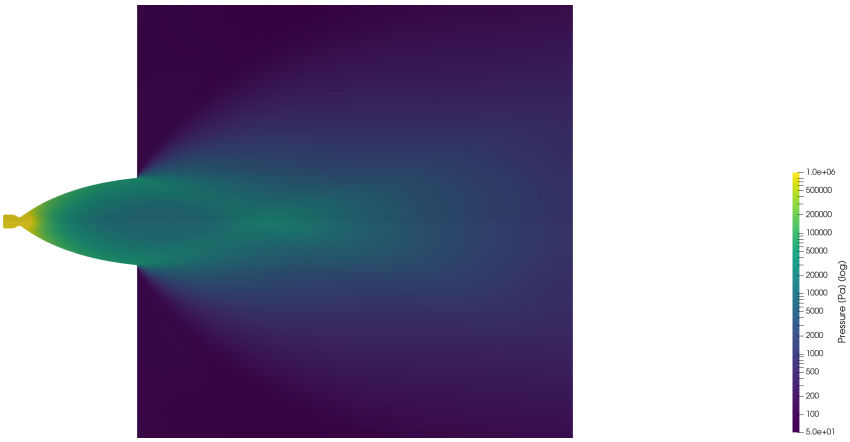


Fig. 5 Stage 2 exhaust plume pressure contours.

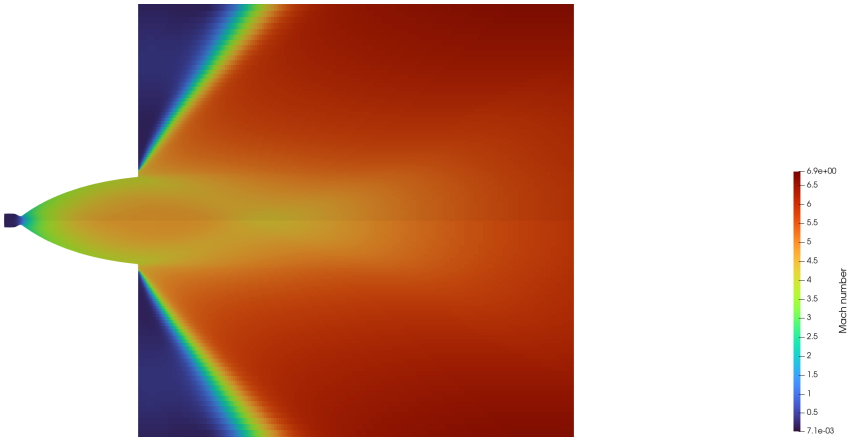


Fig. 6 Stage 2 exhaust plume Mach number contours.

B. Drawings and Models

The following figures show the complete Stage 1 and Stage 2 engine assemblies developed for the COPE vehicle. These assembly models provide the overall configuration and component layout used to support the detailed part drawings that follow.



Fig. 7 Stage 1 engine assembly model.

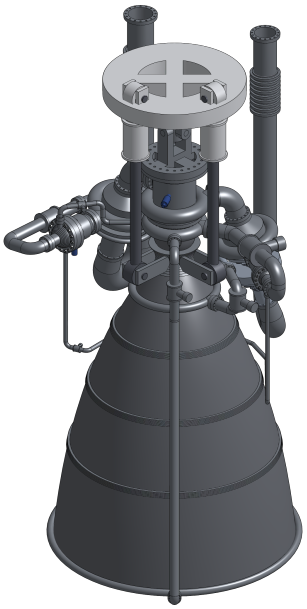
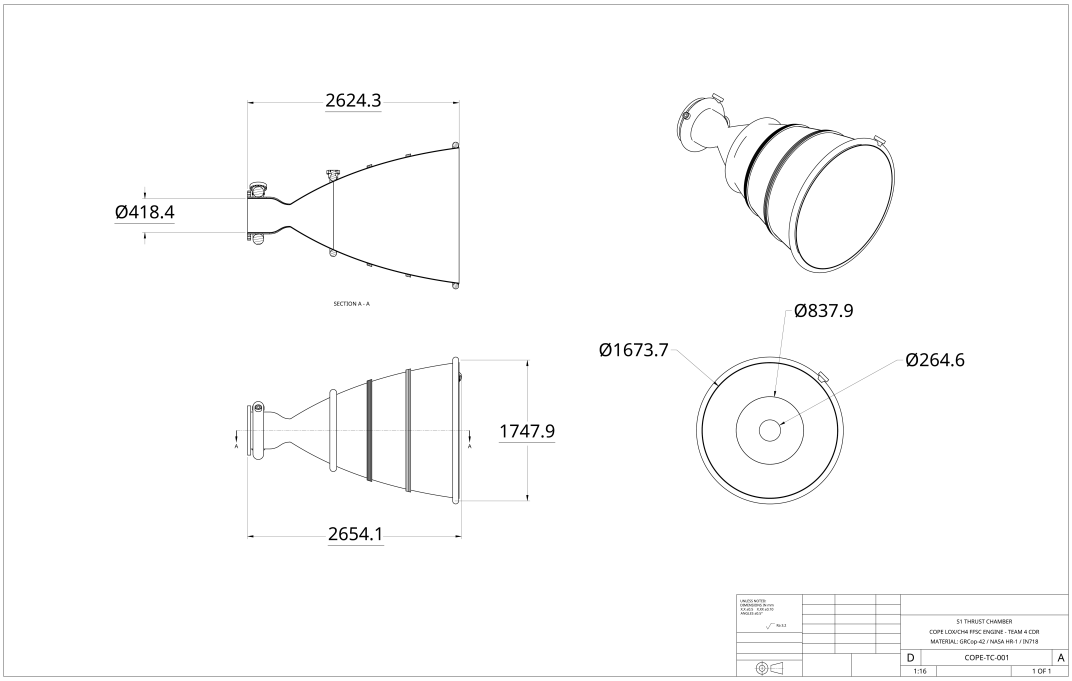


Fig. 8 Stage 2 engine assembly model.

The following drawings provide the detailed dimensions and configurations of the primary propulsion-system components developed for the COPE launch vehicle. Dimensions are provided in millimeters unless otherwise noted on the individual drawing.



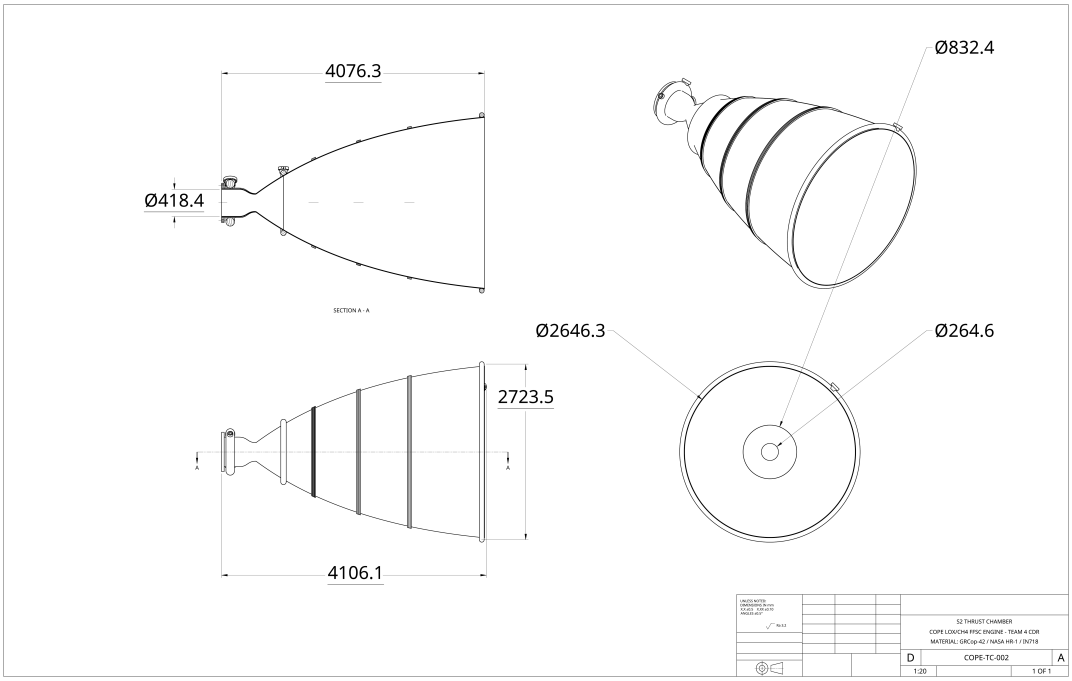
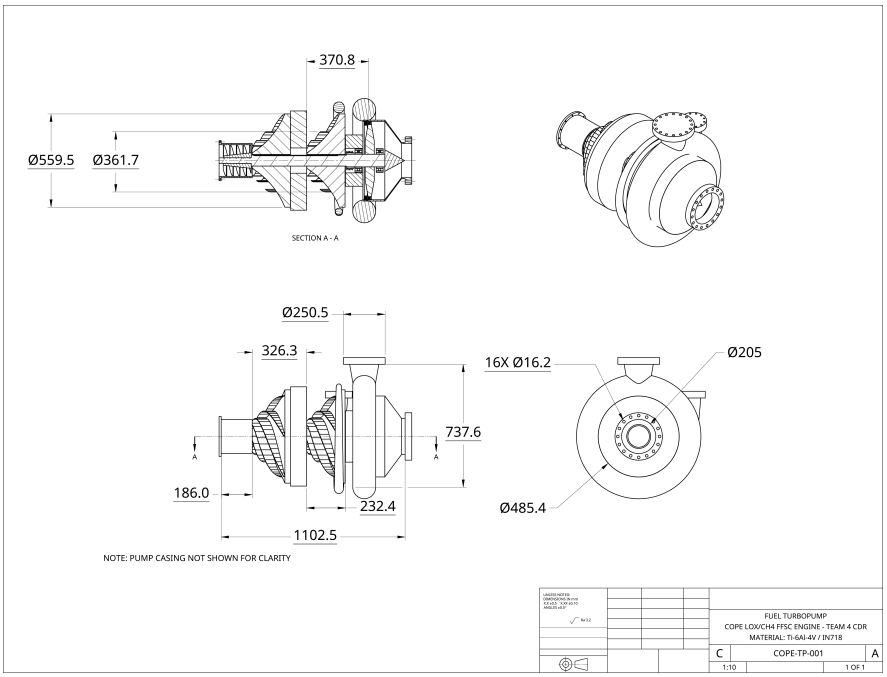


Fig. 9 Stage 1 and Stage 2 thrust chamber detailed drawings.



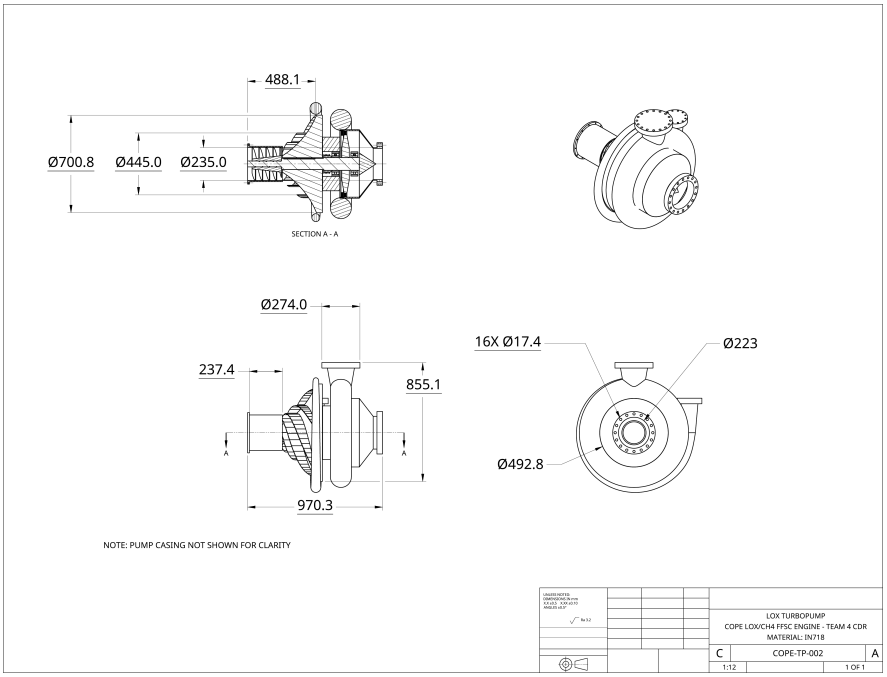
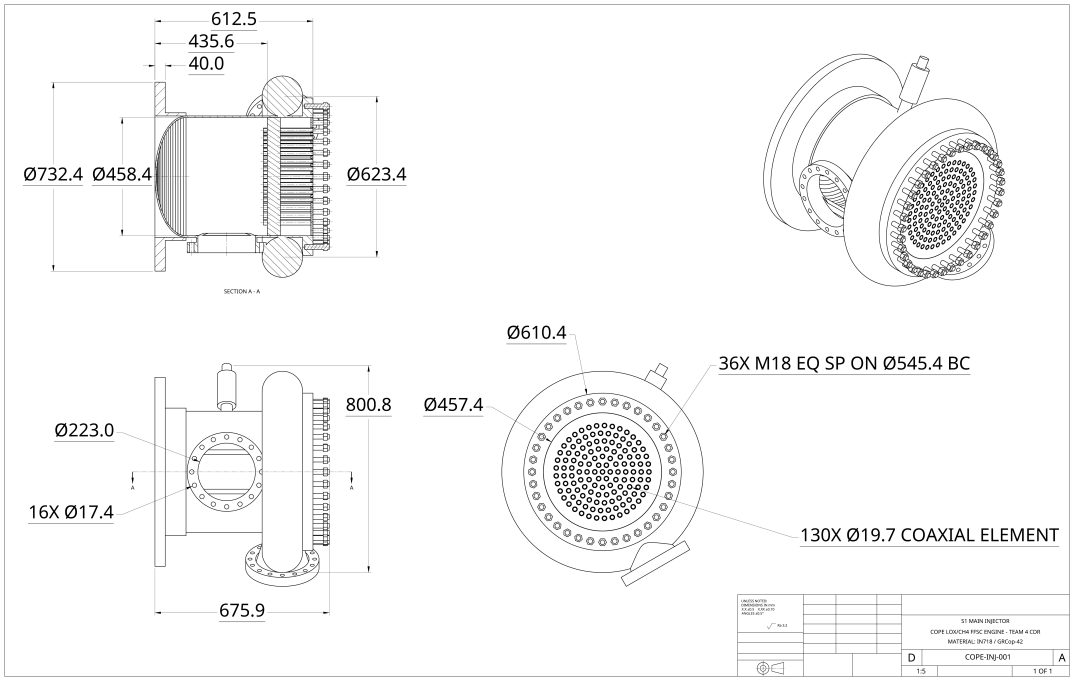


Fig. 10 Fuel and LOX turbopump detailed drawings.



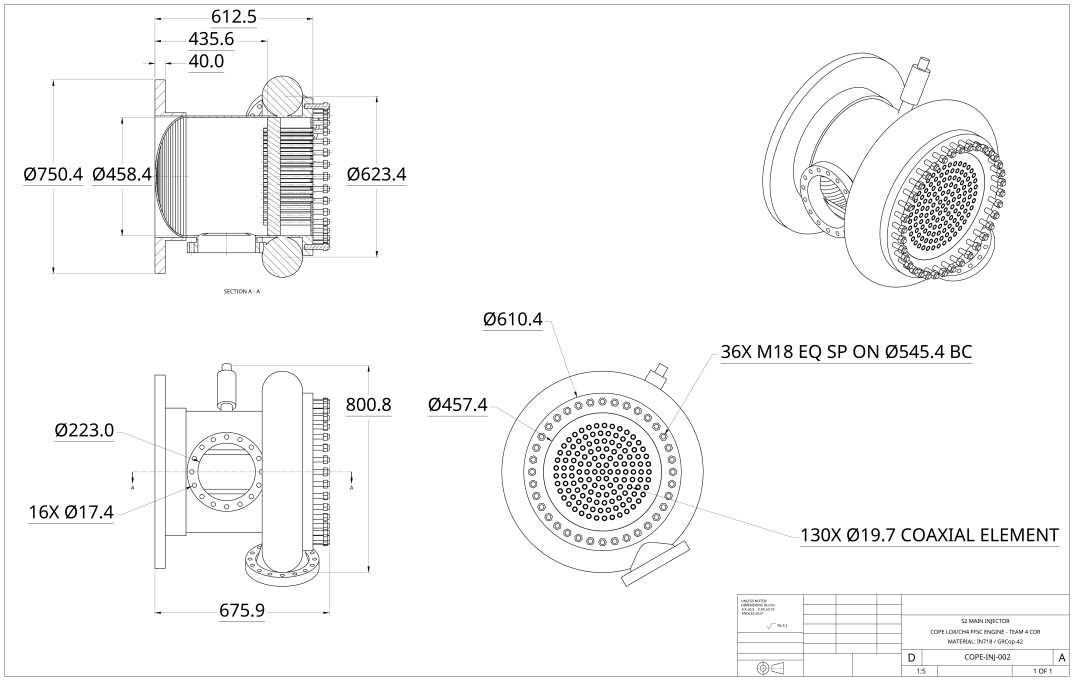
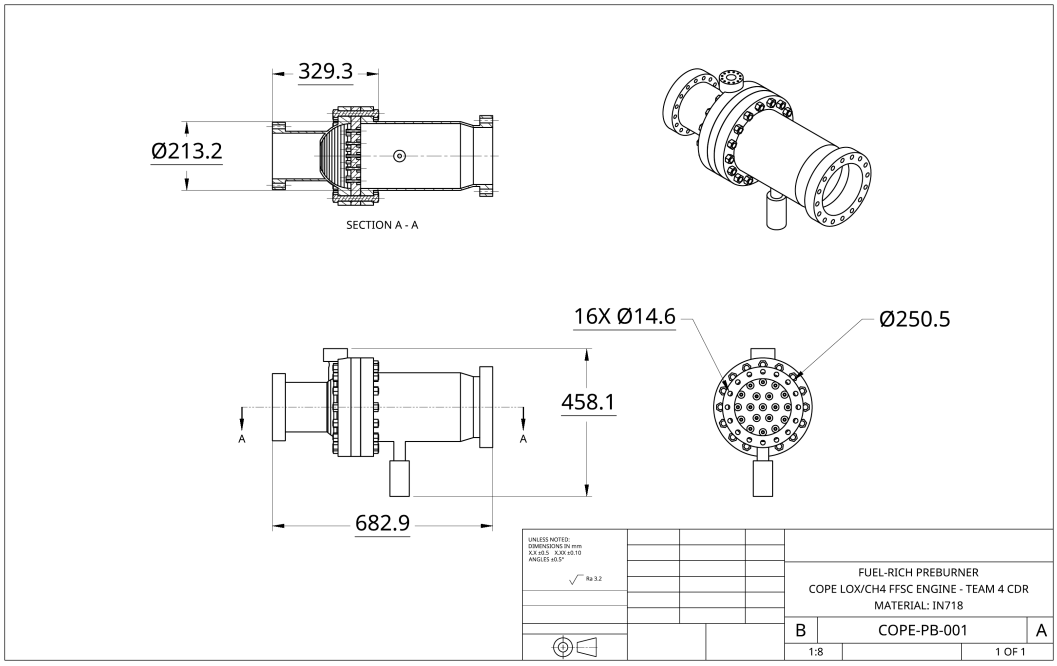


Fig. 11 Stage 1 and Stage 2 main injector detailed drawings.



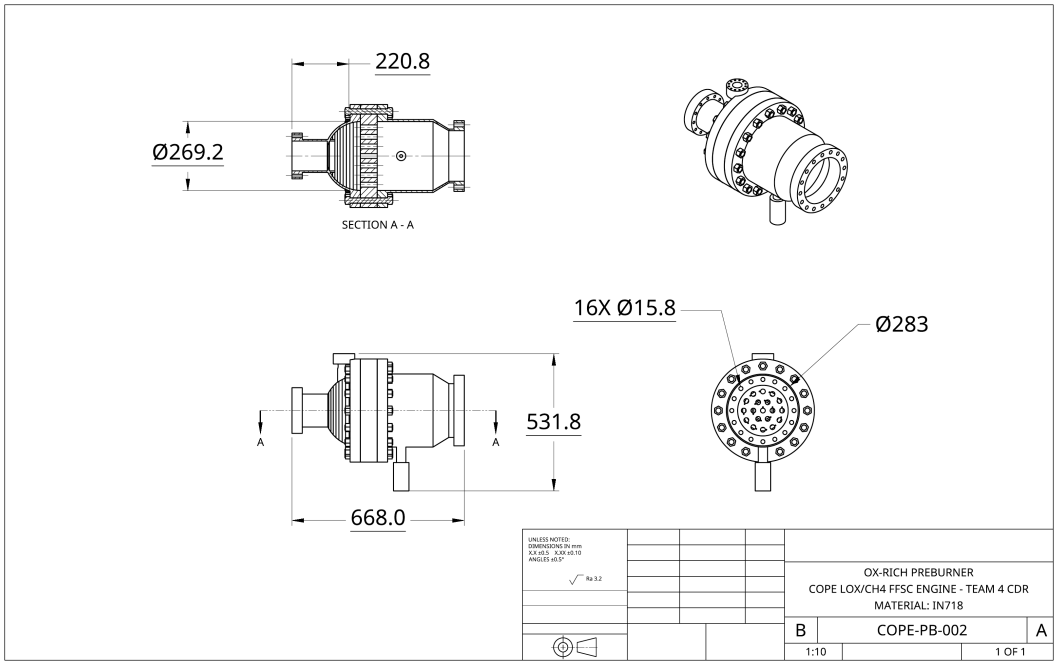
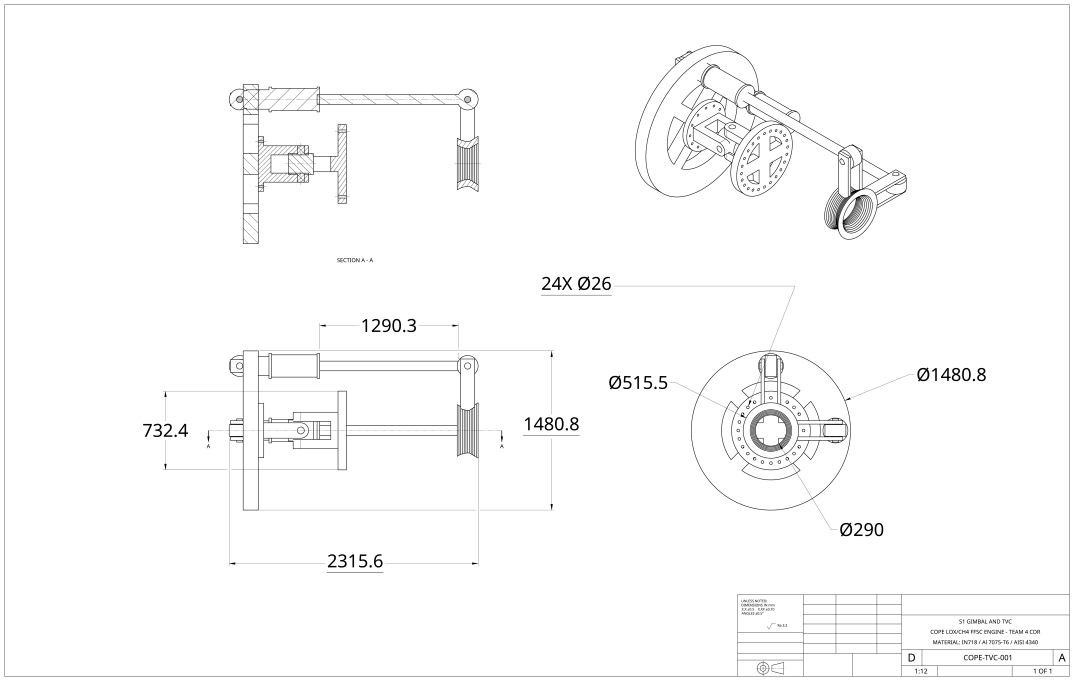


Fig. 12 Fuel-rich and oxidizer-rich preburner detailed drawings.



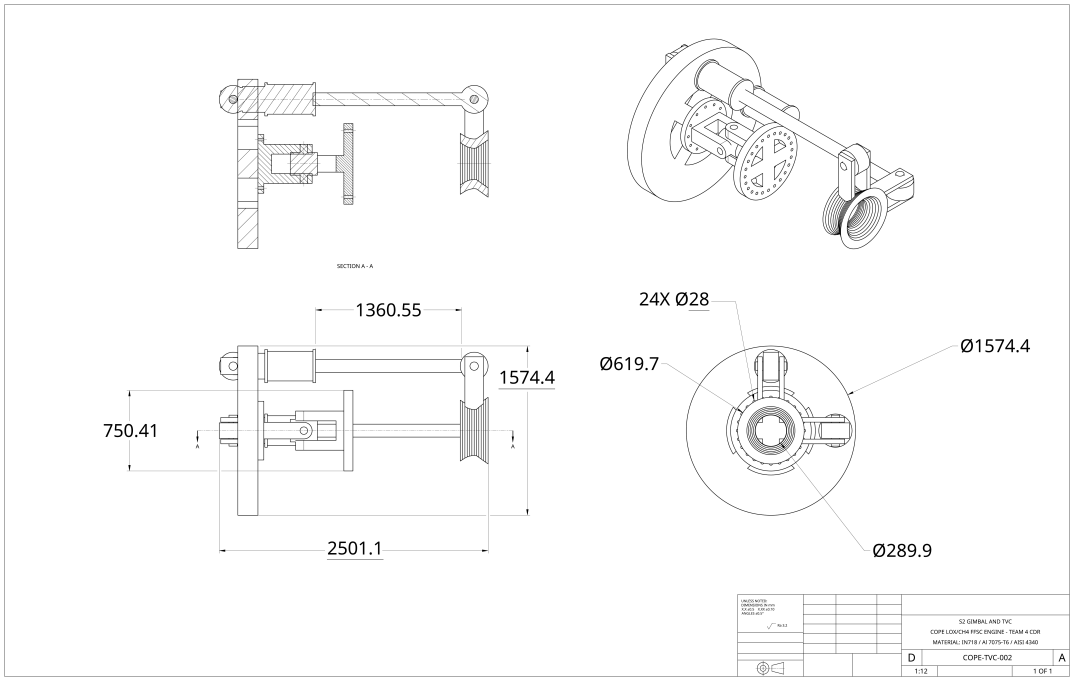


Fig. 13 Stage 1 and Stage 2 gimbal and thrust-vector-control detailed drawings.

C. COTS Part Numbers and Suppliers

Commercial and heritage hardware incorporated into the COPE design is summarized in Table 41. Specific model or part numbers are provided where a component has been identified. Where an exact part number is not available, the selected supplier and applicable hardware family are identified rather than assigning an unsupported part number. Hardware requiring COPE-specific development is identified separately.

Table 41 COTS and Heritage Hardware Suppliers

WBS	Item	Supplier	Model / Part No.	Source
1.2, 1.2.1	Crew capsule and ECLSS	Lockheed Martin / Collins Aerospace	Orion Crew Module with Orion ECLSS class	https://www.lockheedmartin.com/en-us/products/orion.html
1.2.2	Capsule thermal control and TPS	Lockheed Martin / Collins Aerospace	Orion active thermal-control loop and Avcoat heat shield class	https://www.nasa.gov/orion
1.2.3	Launch abort motor	Northrop Grumman	Orion Launch Abort System abort motor class	https://www.northropgrumman.com/space/orion-launch-abort-system
1.2.5	Capsule EPS	Airbus / ESA	Orion European Service Module solar arrays and batteries class	https://www.esa.int/Science_Exploration/Human_and_Robotic_Exploration/Orion
1.2.6	Capsule RCS thruster	Aerojet Rocketdyne / L3Harris	MR-107 class monopropellant thruster, 110 N; MR-104 class for larger maneuvers	https://www.l3harris.com/all-capabilities/monopropellant-rocket-engines
1.1.1	Sample cryocooler	Northrop Grumman	High Efficiency Cryocooler (HEC) class pulse-tube cooler	https://www.northropgrumman.com/space/cryocoolers
1.3.1.3 / 1.4.1.3 / 1.5.1.3	TVC actuator	Moog Inc., Space and Defense	Modular Electromechanical Actuator (MEMA) family, Form 500-1022	https://www.moog.com/content/dam/moog/literature/sdg/space/actuationmechanisms/moog-ElectromechanicalActuators-Datasheet.pdf
1.3.1.3 / 1.4.1.3	TVC actuation controller	Moog Inc., Space and Defense	Launch-vehicle TVC actuation controller family	https://www.moog.com/products/controllers-drives-software/actuation-controllers-for-launch-vehicle-booster-and-missile.html
1.3.1.1 / 1.4.1.1	Chamber and preburner pressure transducers	Kulite Semiconductor Products	XTEH-10L-190; HKM-375 series for tank and line pressure	https://kulite.com/
1.3.1.2 / 1.4.1.2	Cryogenic solenoid and pilot valves	Marotta Controls	CoRe MV100X and MV524X cryogenic valve families	https://marotta.com/products/flow-controls/pressure-controls/cryogenic-valves/
1.3.1.2 / 1.4.1.2	Pressurization regulators	Marotta Controls	Space-qualified pressure regulator / reducer family	https://marotta.com/products/flow-controls/pressure-controls/pressure-regulators-reducers/

WBS	Item	Supplier	Model / Part No.	Source
1.3.1.2 / 1.4.1.2	Stage prevalves	Valcor Engineering	Cryogenic launch-vehicle ball / butterfly valve family	https://www.valcor.com/cryogenic-launch-vehicle-valves/
1.3.1.1 / 1.4.1.1	Igniter and exciter	Unison Industries / GE Aerospace	Solid-state ignition exciter family with augmented spark igniter torch	https://www.unisonindustries.com/electrical/ignition
1.3.1.2 / 1.4.1.2	Flex joints and bellows	Senior Aerospace Bird Bellows	Gimbal-compensating flexible metal bellows for 259 / 227 mm feed lines	https://www.seniorplc.com/about-us/operating-businesses/aerospace/bird-bellows.aspx
1.3.5 / 1.5.5	Stage separation joint	Ensign-Bickford Aerospace & Defense	Frangible Joint assembly	https://ebad.com/
1.3.5 / 1.5.5	Separation pushers	Ensign-Bickford Aerospace & Defense	Separation Bolt series and pneumatic / pyrotechnic pusher hardware	https://ebad.com/products/separation-bolt-series/
1.3.5 / 1.2.3	Ordnance initiators	Pacific Scientific Energetic Materials Company	NASA Standard Initiator (NSI) class initiators and pyro valves	https://psemc.com/products/rocket-motor-igniter/
1.3.6 / 1.4.6 / 1.5.6	Stage RCS thruster	Aerojet Rocketdyne / L3Harris	MR-107 class monopropellant thruster, 110 N	https://www.l3harris.com/all-capabilities/monopropellant-rocket-engines
1.3.3 / 1.4.3 / 1.5.3	Cryogenic tank insulation	Dunmore	Double-Aluminized Kapton (DAK) and embossed DAK multilayer-insulation films	https://www.dunmore.com/products/double-aluminized-polyimide-film.html
1.3.1.3 / 1.4.4 / 1.5.4	Vehicle batteries	Saft	VL51ES Generation 6 high-rate space Li-ion cells	https://www.directindustry.com/prod/saft/product-19840-2536062.html
1.3.7 / 1.4.7 / 1.5.7 / 1.2.7	Flight computer and fault detection	Moog Inc., Space and Defense	Integrated Avionics Unit (IAU) / Main Avionics Controller family	https://www.moog.com/content/dam/moog/literature/sdg/space/avionics/moog-integrated-avionics-unit-datasheet.pdf
1.4.5 / 1.5.5	Cryogenic transfer coupling	Development Item	Ship-to-ship cryogenic quick disconnect, 100 kg/s class	https://www.nasa.gov/fluid-transfer/
1.6	Launch pad and mobile launcher	NASA Kennedy Space Center	LC-39B class pad and mobile launcher	https://www.nasa.gov/kennedy
1.3.1.1 / 1.4.1.1	Engine main and preburner valves	Aerojet Rocketdyne / Moog	High-pressure cryogenic MFV, MOV, FPOV, FPFV, OPFV, and CCV heritage class	https://www.moog.com/content/dam/moog/literature/sdg/space/propulsion/moog-cryogenic-ball-valves-datasheet.pdf
1.3.1.2 / 1.4.1.2	Turbopump hybrid bearings	Timken / Barden	Hybrid angular-contact bearings with 440C races and Si_3N_4 balls	https://www.timken.com/markets/aerospace-defense/

WBS	Item	Supplier	Model / Part No.	Source
1.3.1.3 / 1.4.1.3 / 1.4.4	TVC battery	Moog Inc., Space and Defense	Reconfigurable Integrated Battery (RIB), High Power, 48S3P	https://www.moog.com/products/power-distribution-management/reconfigurable-integrated-battery.html
1.5.6 / 1.4.5	Docking and refueling interface mechanism	NASA / Boeing	NASA Docking System / IDSS androgynous soft-capture docking mechanism	https://www.internationaldockingstandard.com/
1.1.1 / 1.3.3 / 1.4.3 / 1.5.3	Cryogenic temperature sensor	Lake Shore Cryotronics	DT-670-SD silicon diode temperature sensor	https://www.lakeshore.com/products/categories/overview/temperature-products/cryogenic-temperature-sensors/dt-670-silicon-diodes
1.3.1.2 / 1.4.1.2	Auxiliary fluid safety screen	The Lee Company	HI-BAR safety-screen family	https://www.theleeco.com/industries/space/products/safety-screens/
1.2.6 / 1.4.6 / 1.5.6	Inertial attitude measurement unit	Honeywell	Miniature Inertial Measurement Unit (MIMU)	https://www.honeywellaerospace.com/us/en/products-and-services/products/navigation-and-sensors/inertial-measurement-units/miniature-inertial-measurement-unit
1.2.6 / 1.4.6 / 1.5.6	Star tracker	Sodern	Hydra	https://sodern.com/en/ranges/hydra
1.2.2 / 1.3.3 / 1.4.3 / 1.5.3	Flexible survival heater	Minco	Space-flight polyimide flexible-heater class	https://www.minco.com/products/flexible-heaters/

The following figures illustrate the commercial and heritage hardware identified for the COPE vehicle.



Fig. 14 Lockheed Martin Orion crew module and ECLSS.

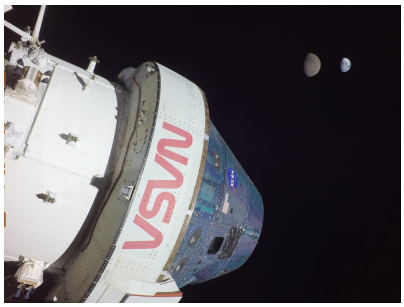


Fig. 15 Lockheed Martin Orion active thermal control and Avcoat heat shield.



Fig. 16 Northrop Grumman Orion Launch Abort System abort motor.

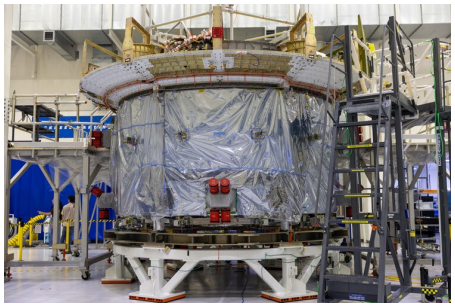


Fig. 17 Airbus / ESA Orion European Service Module solar arrays and batteries.



Fig. 18 Aerojet Rocketdyne MR-107 monopropellant thruster, capsule RCS.

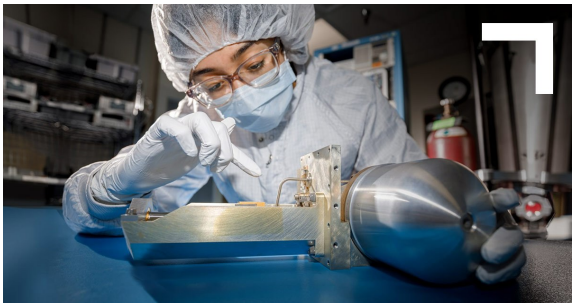


Fig. 19 Northrop Grumman High Efficiency Cryocooler, sample storage.

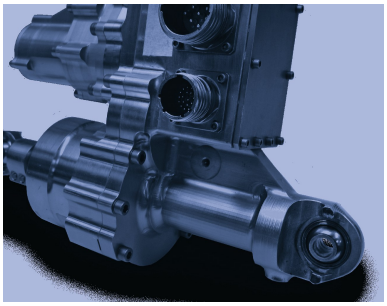


Fig. 20 Moog Modular Electromechanical Actuator (MEMA), TVC actuator.



Fig. 21 Moog launch-vehicle TVC actuation controller.

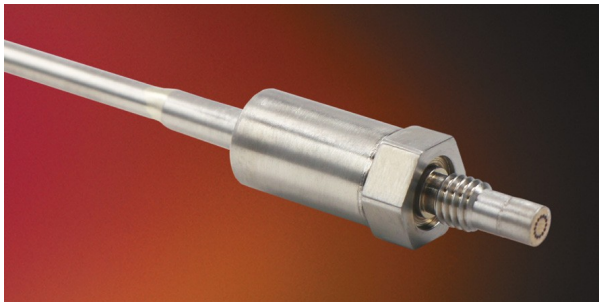


Fig. 22 Kulite XTEH-10L-190 high-temperature pressure transducer.

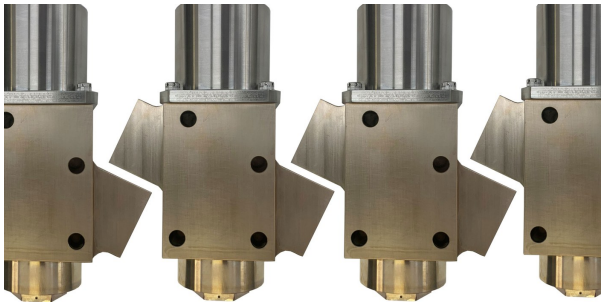


Fig. 23 Marotta Controls CoRe MV100X cryogenic solenoid valves.



Fig. 24 Valcor Engineering cryogenic launch-vehicle prevalve.



Fig. 25 Unison Industries solid-state ignition exciter.



Fig. 26 Senior Aerospace Bird Bellows gimbal-compensating feed-line flex joint.



Fig. 27 Ensign-Bickford Aerospace & Defense frangible joint assembly.



Fig. 28 Ensign-Bickford Aerospace & Defense separation bolt and pusher hardware.



Fig. 29 Pacific Scientific EMC NASA Standard Initiator.



Fig. 30 Aerojet Rocketdyne MR-107 monopropellant thruster, stage RCS.



Fig. 31 Dunmore Double-Aluminized Kapton multilayer insulation.



Fig. 32 Saft VL51ES lithium-ion cells, vehicle batteries.



Fig. 33 Moog Integrated Avionics Unit, flight computer and fault detection.



Fig. 34 NASA Kennedy Space Center Launch Complex 39B.



Fig. 35 Timken / Barden hybrid angular-contact turbopump bearing.

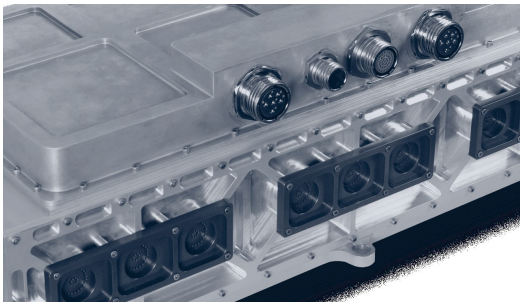


Fig. 36 Moog Reconfigurable Integrated Battery, TVC power.



Fig. 37 Marotta Controls space-qualified pressure regulator.



Fig. 38 Moog high-pressure cryogenic ball valve, engine main and preburner valves.

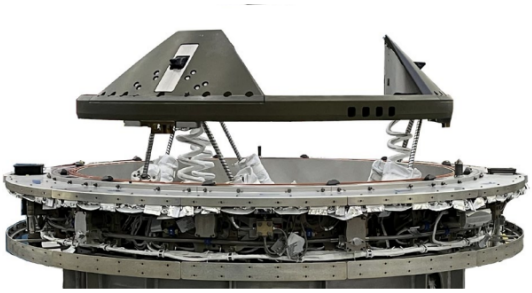


Fig. 39 NASA Docking System / IDSS soft-capture docking mechanism.

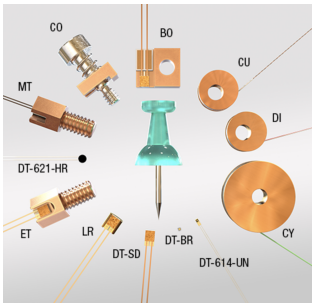


Fig. 40 Lake Shore Cryotronics DT-670-SD silicon diode temperature sensor.



Fig. 41 The Lee Company HI-BAR safety screen.



Fig. 42 Honeywell Miniature Inertial Measurement Unit.



Fig. 43 Sodern Hydra star tracker.



Fig. 44 Minco polyimide flexible survival heater.